

GA102 DT SKU P-BOARD
350W, FH Std PCB, 384b, GDDR6X x8
DP + DP + DP + HDMI/DP

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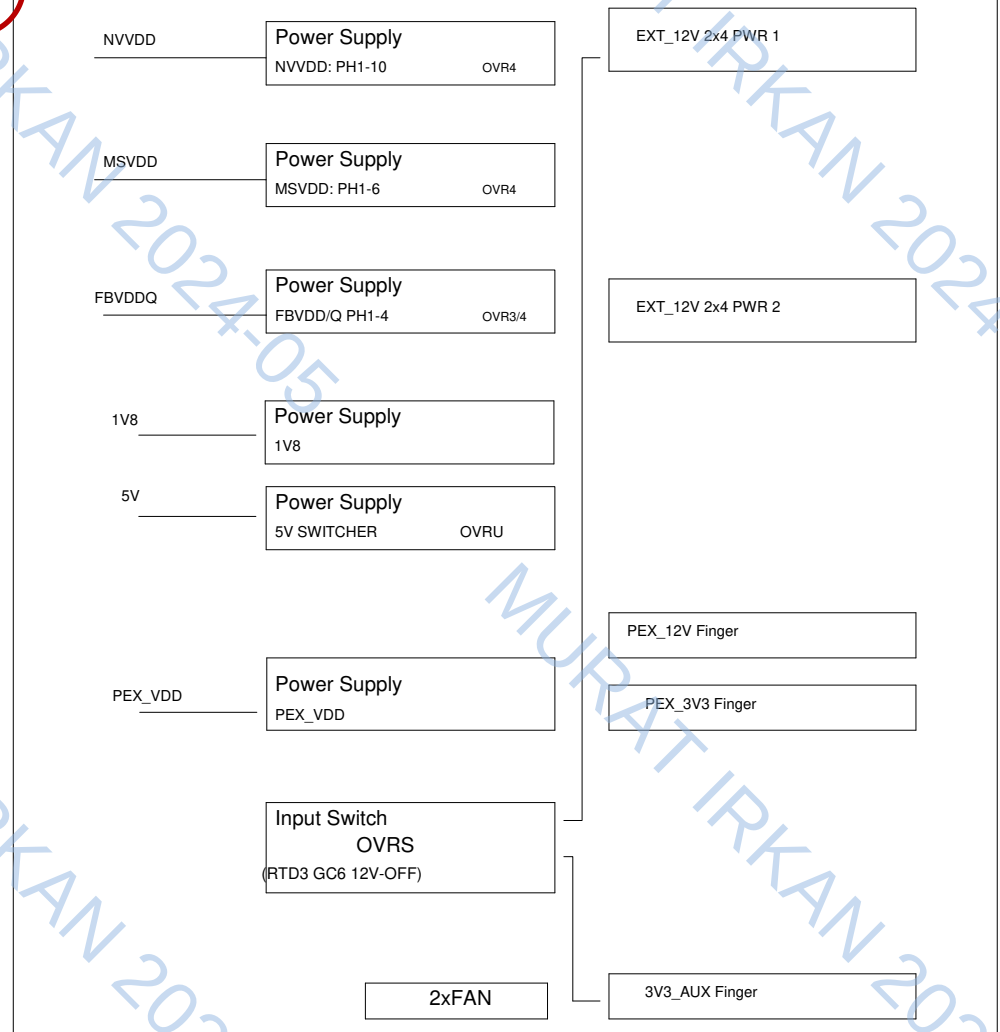
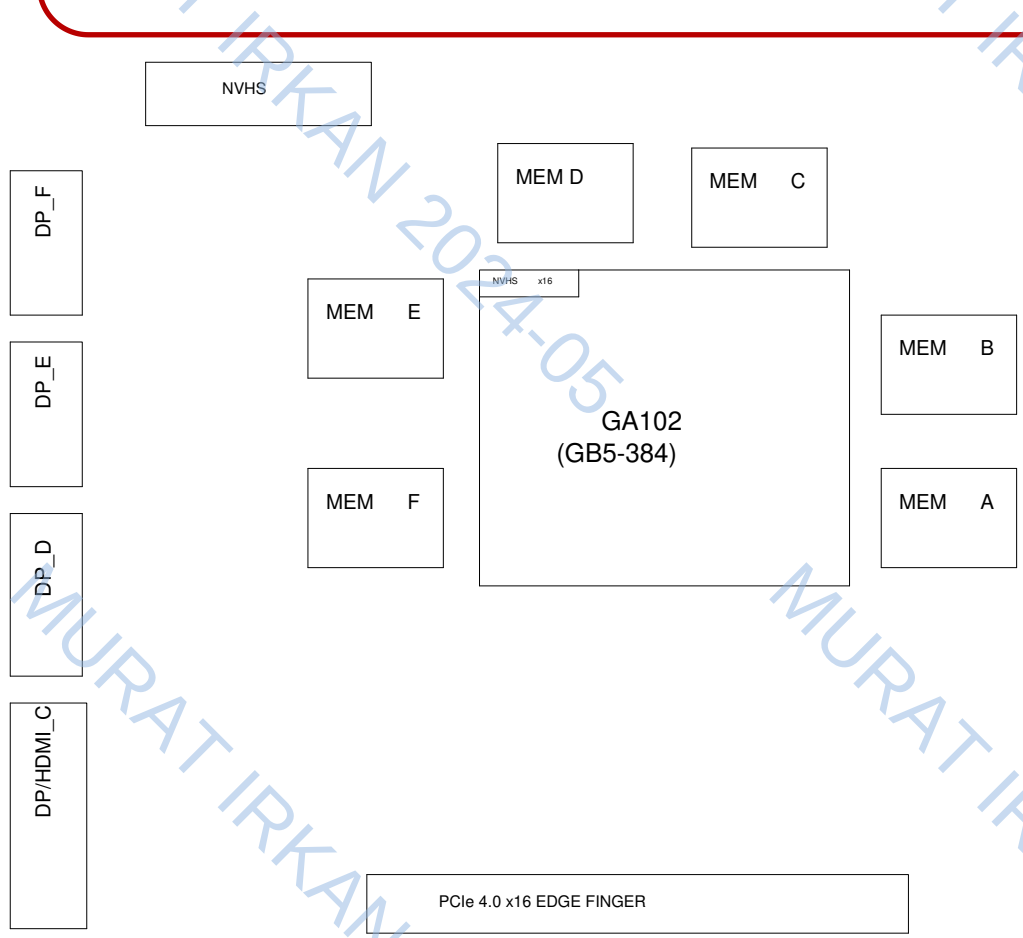
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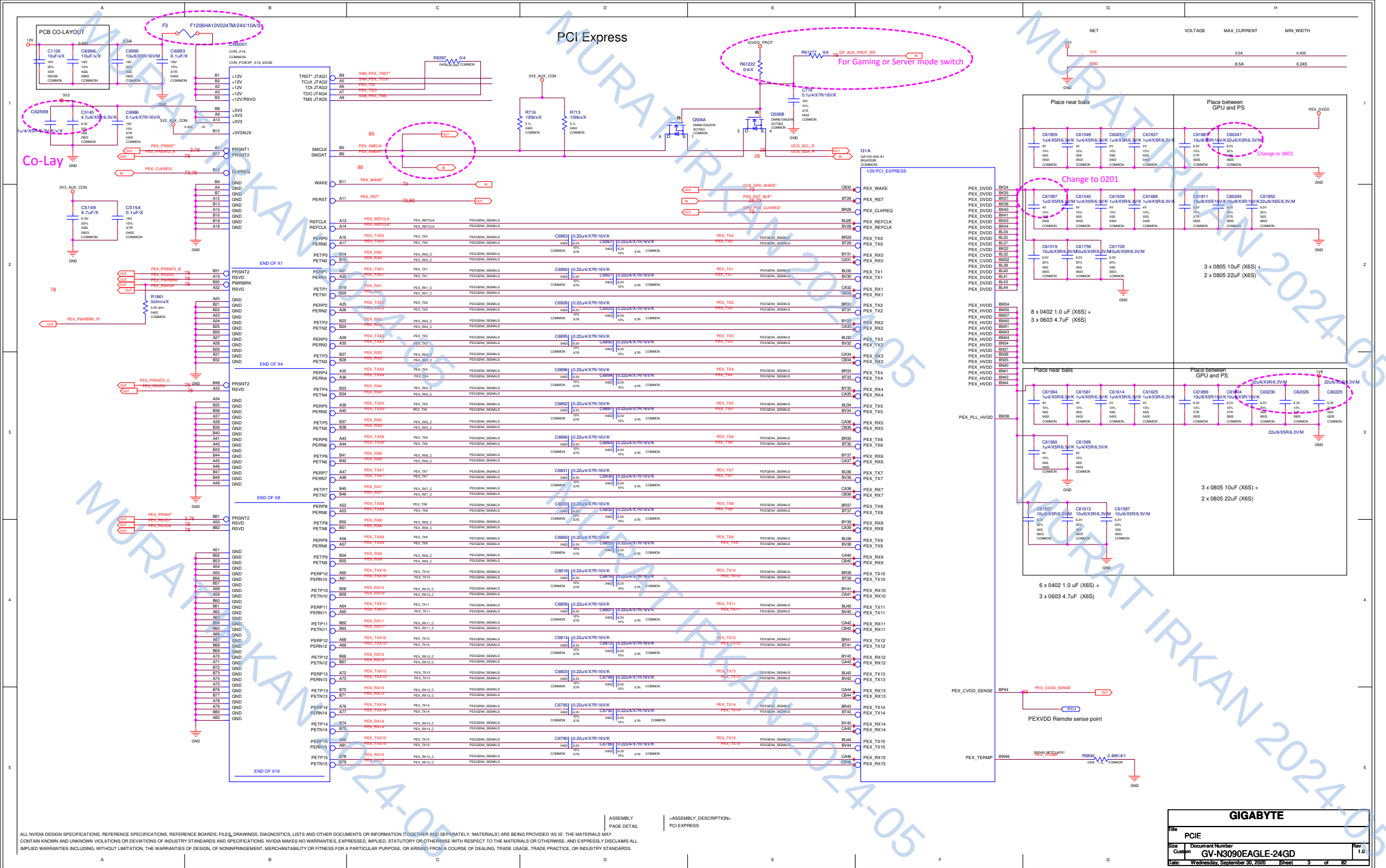
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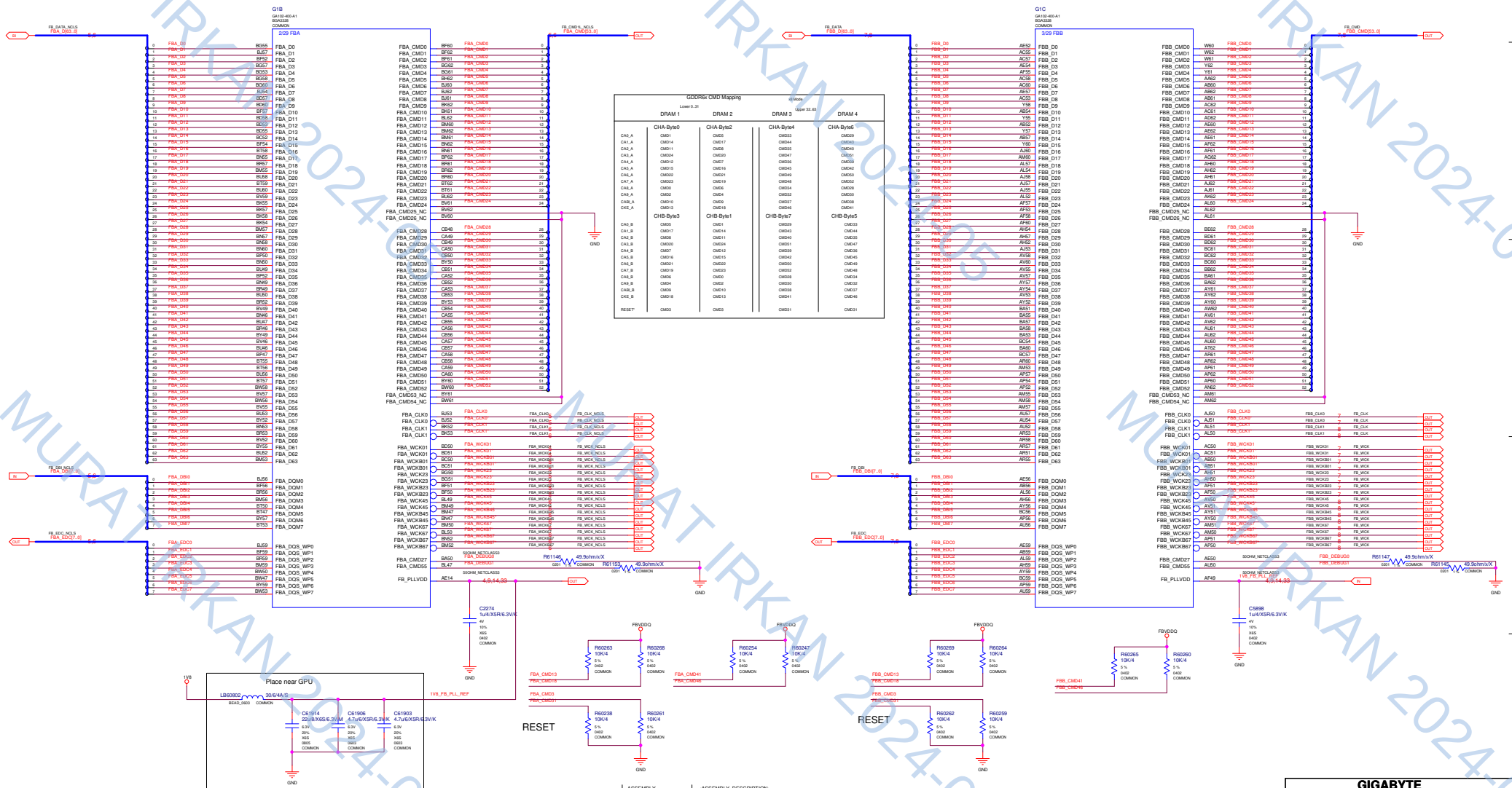


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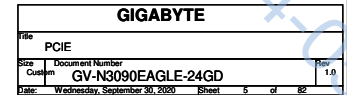
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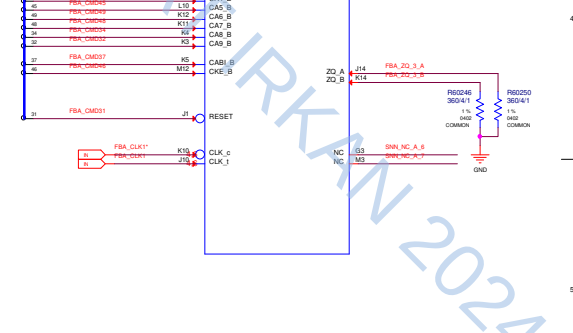
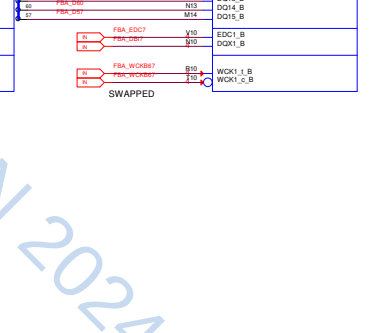
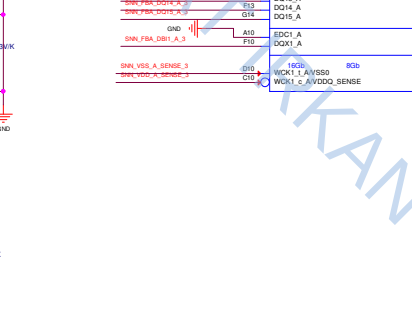
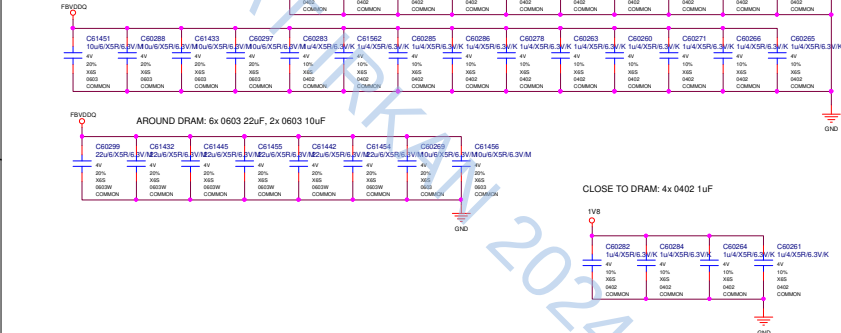
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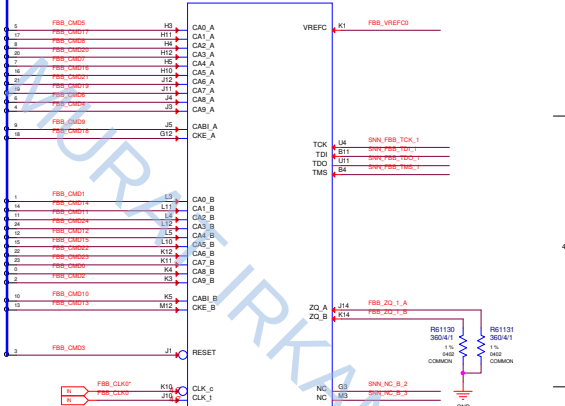
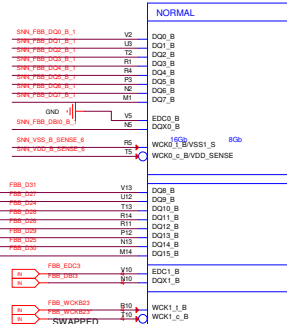
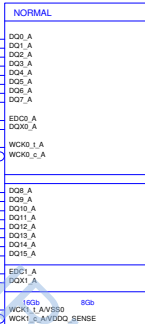
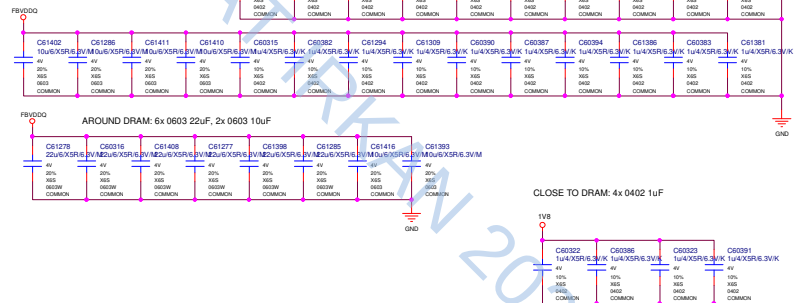


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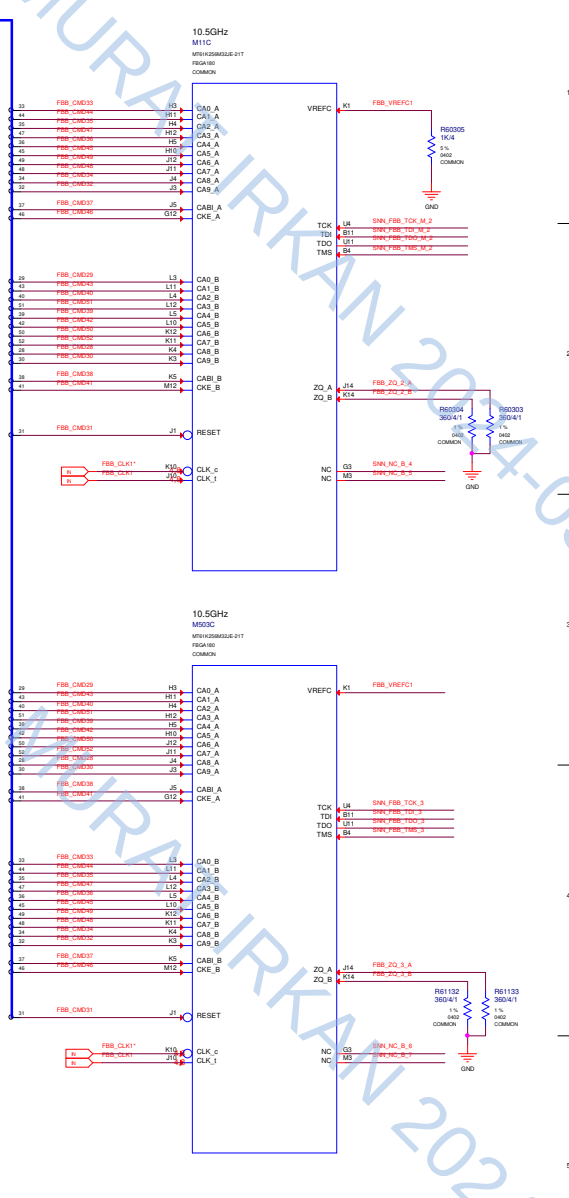
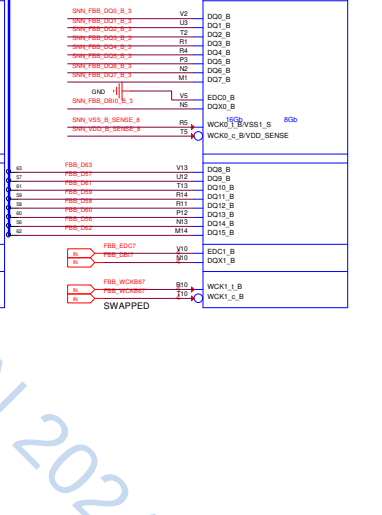
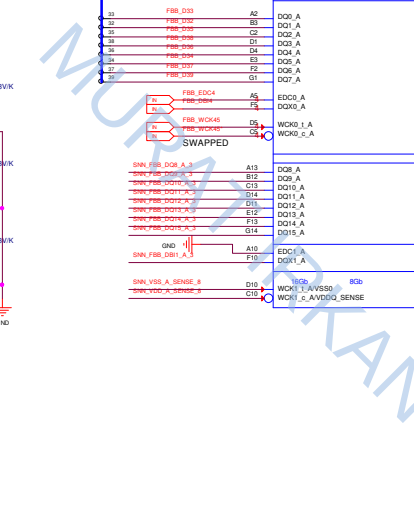
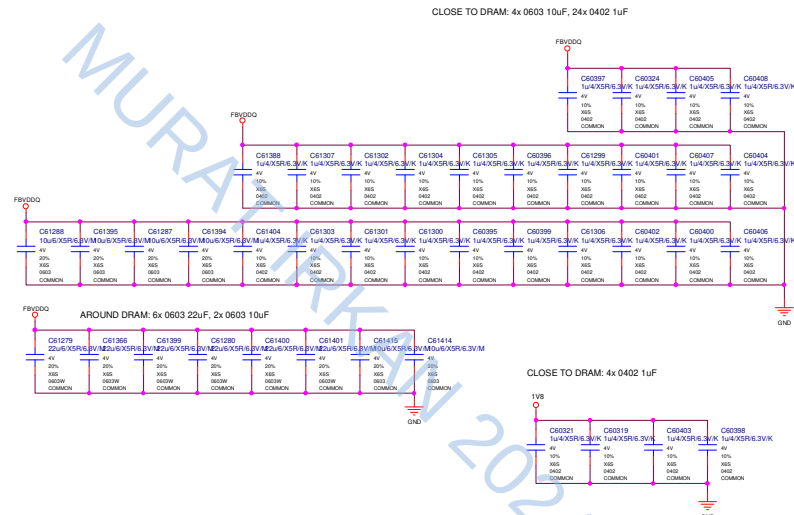




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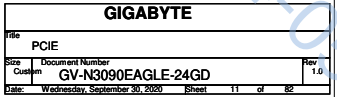


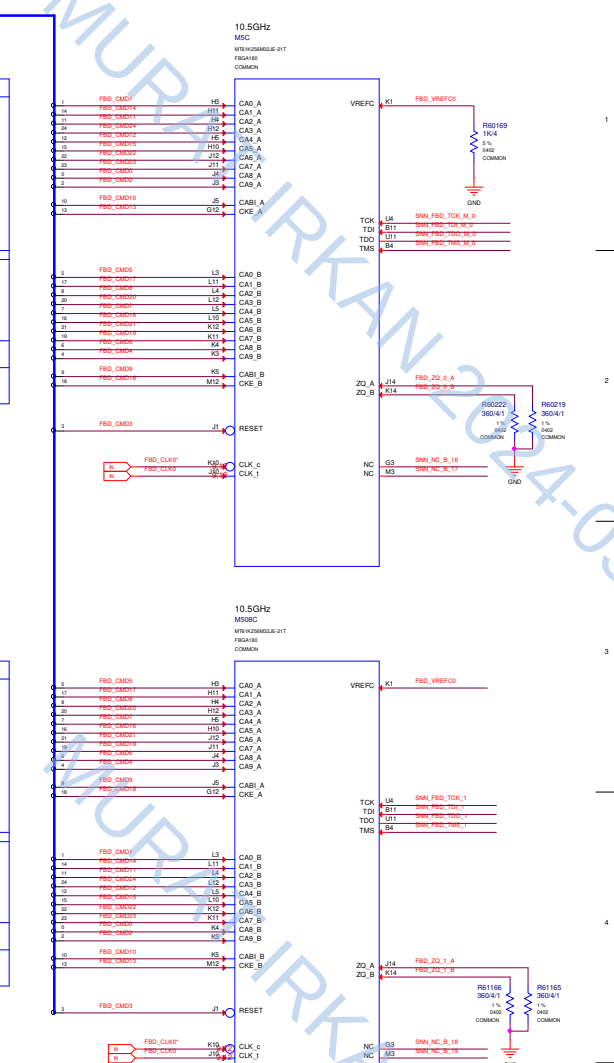
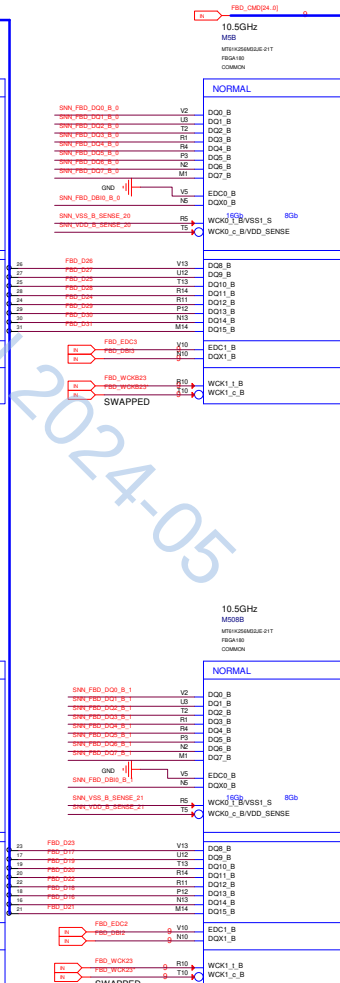
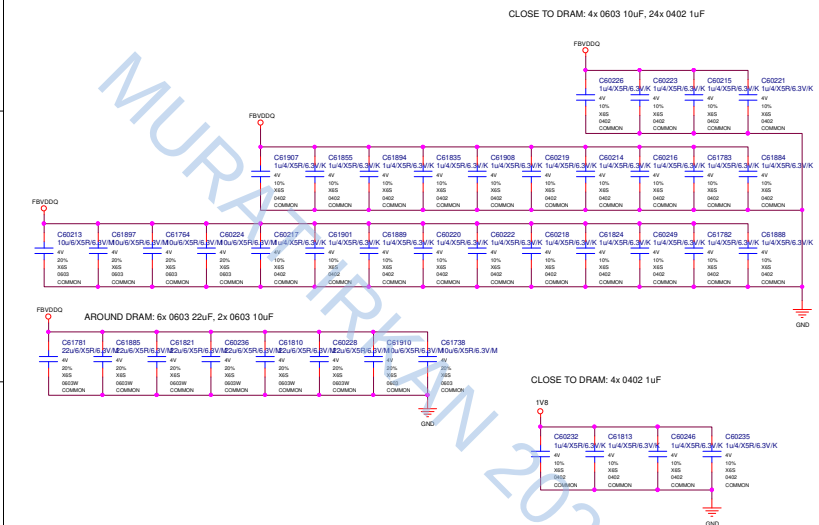
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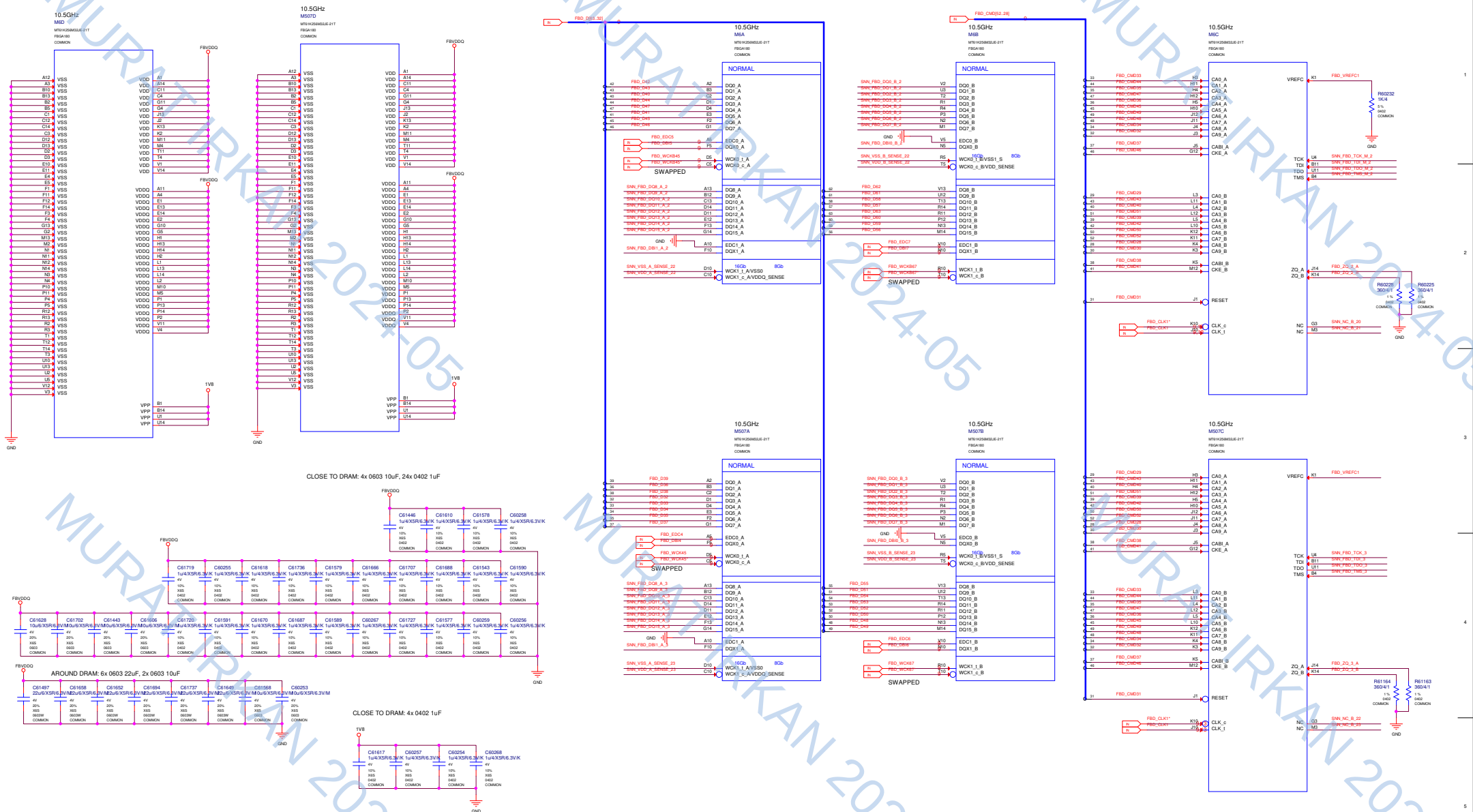




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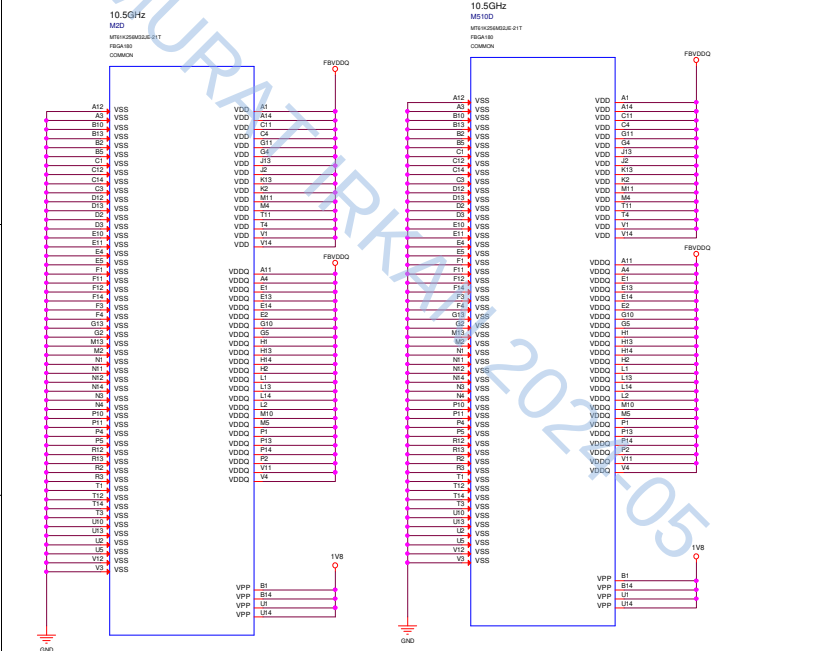




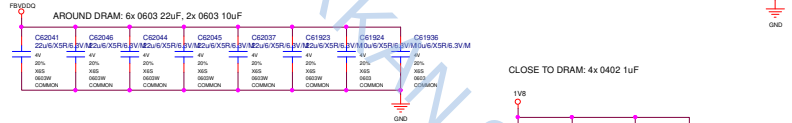
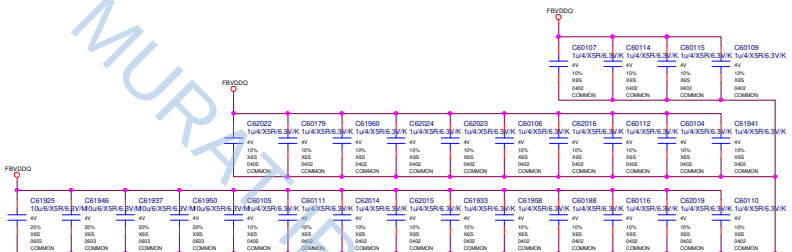
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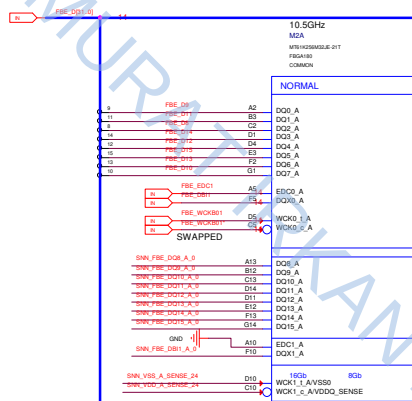
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PCIE			
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CLOSE TO DRAM: 4x 0603 10uF, 24x 0402 1uF



CLOSE TO DRAM: 4x 0402 1uF



10.5GHz

M10D

M10B

M10C

M10D

M10B

M10C

M10D

M10B

M10C

M10D

M10B

M10C

M10D

M10B

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M10B

M10C

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M10B

M10C

M10D

M10B

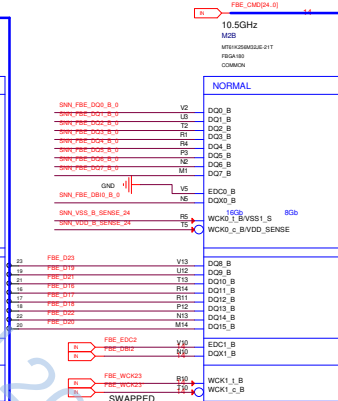
M10C

M10D

M10B

M10C

M10D



10.5GHz

M10D

M10B

M10C

M10D

M10B

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M10B

M10C

M10D

M10B

M10C

M10D

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M10C

M10D

M10B

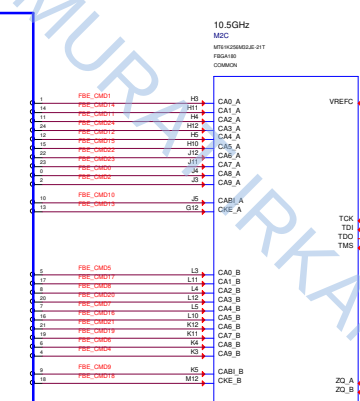
M10C

M10D

M10B

M10C

M10D



10.5GHz

M10D

M10B

M10C

M10D

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M10B

M10C

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M10C

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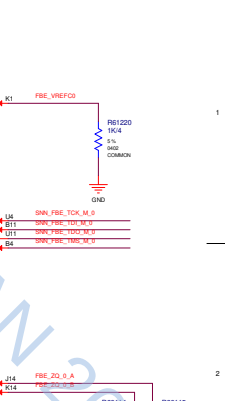
M10C

M10D

M10B

M10C

M10D



10.5GHz

M10D

M10B

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M10D

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M10B

M10C

M10D

M10B

M10C

M10D

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M10C

M10D

M10B

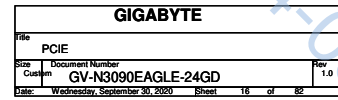
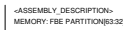
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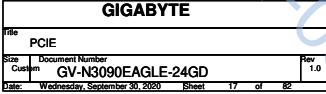
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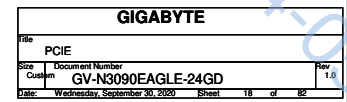
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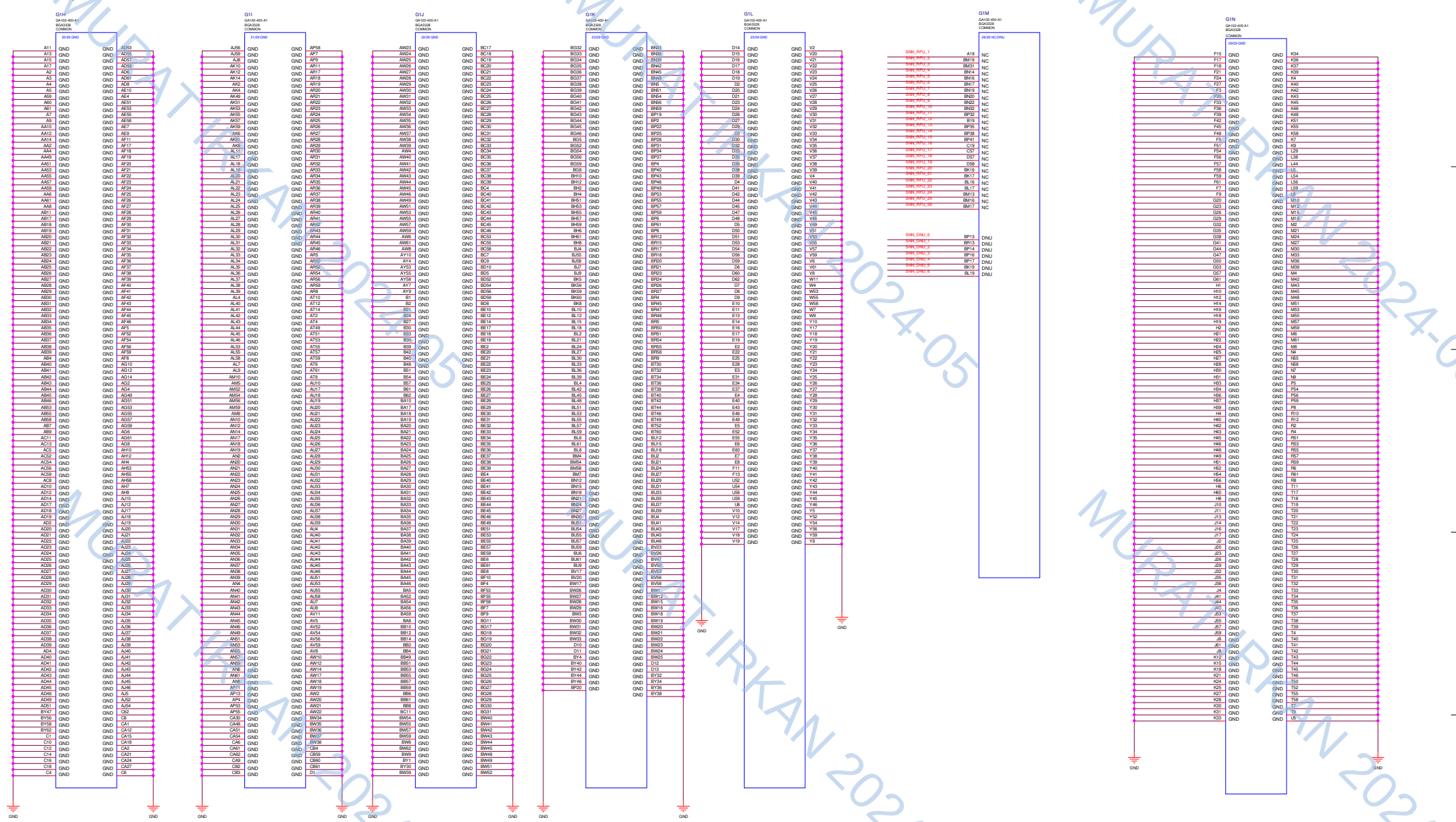
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MEMORY FBE PARTITION 31..0

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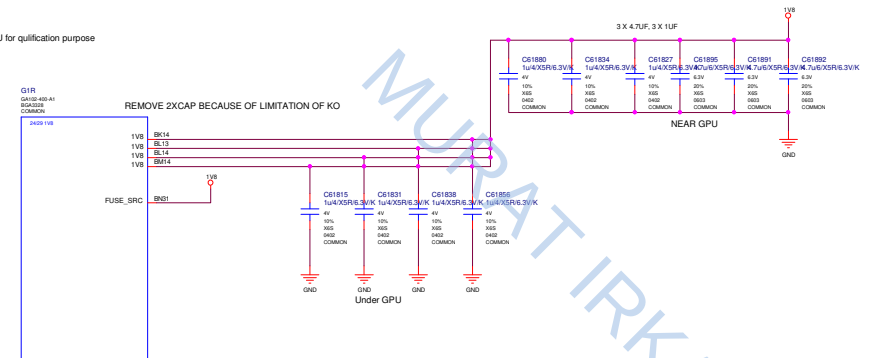
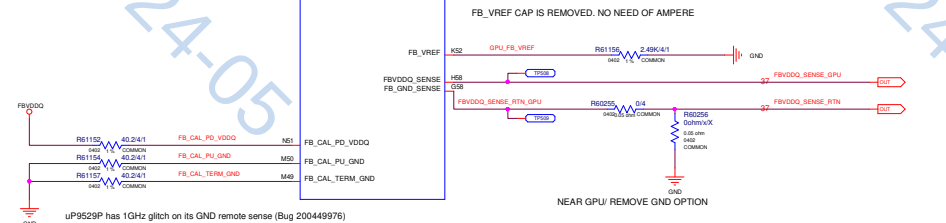
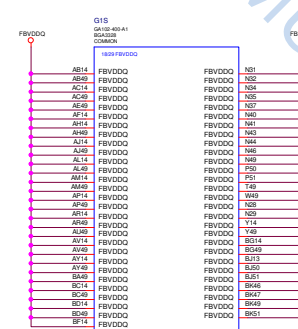
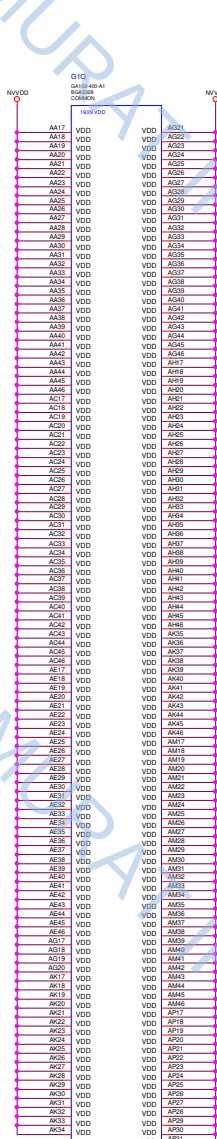
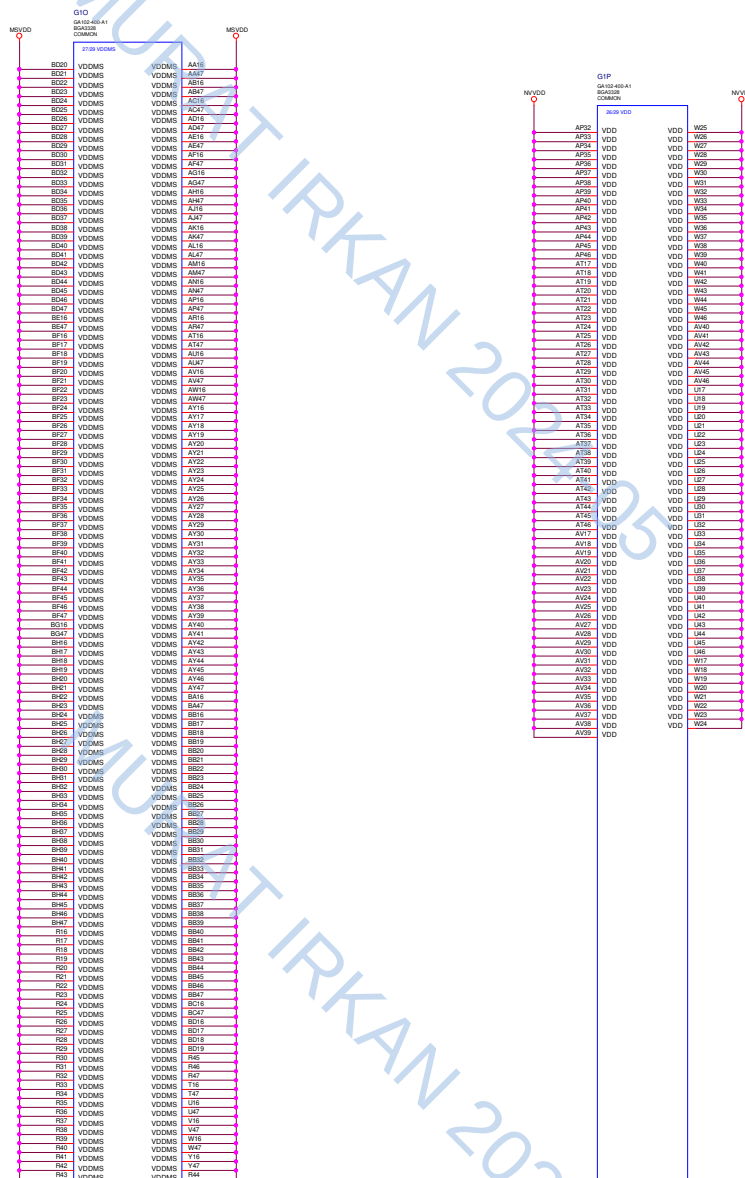
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GPU GND, RFUs & RSVD

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GIGABYTE

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GPU PWR and GND



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GPU POWERS

GIGABYTE

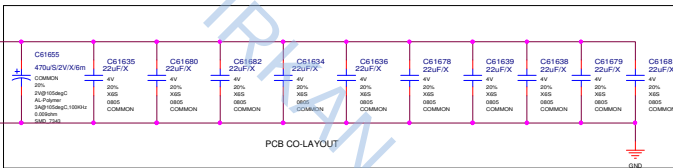
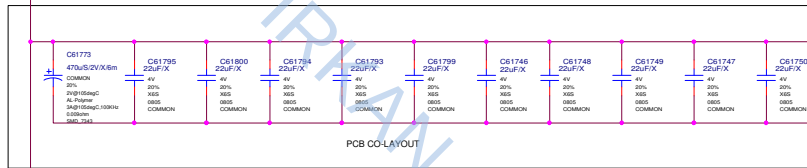
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Size	Document Number		Rev
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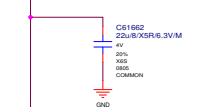
NVVDD

UNDER GPU

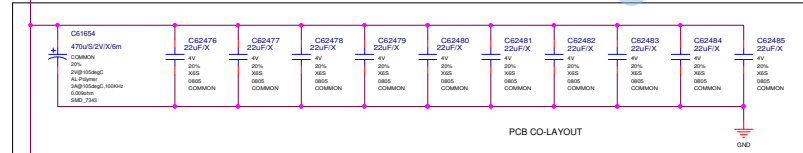
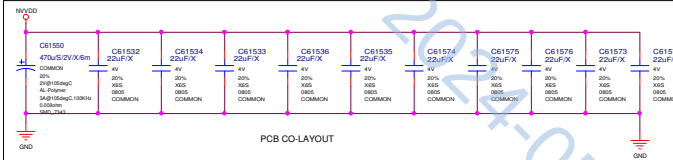
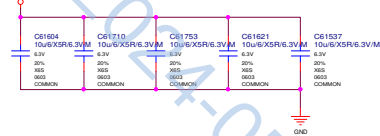
4x SMD7343 220uF (EACH CO-LAYOUT WITH 10x 0805 22uF)



1x 0805 22uF

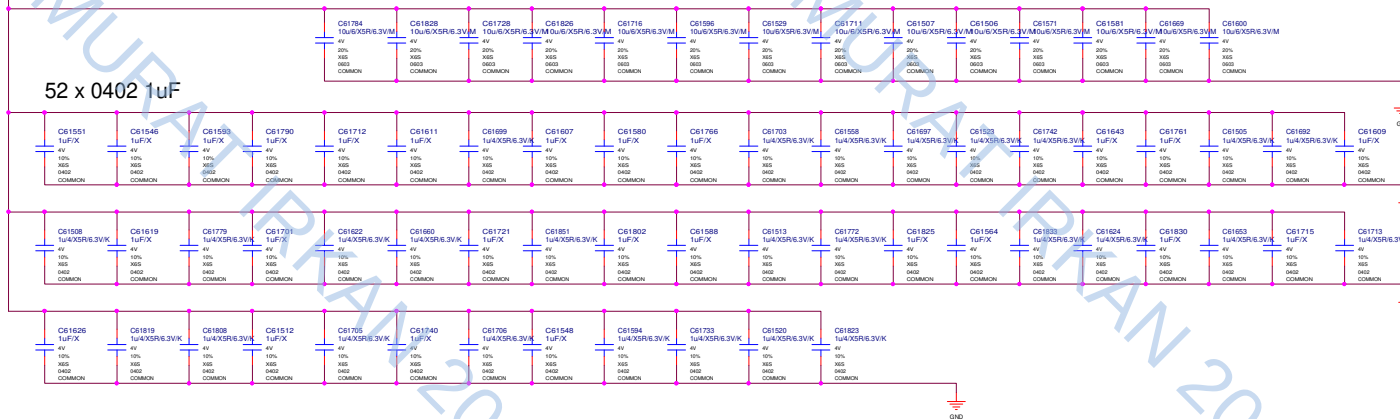


5x 0603 4.7uF

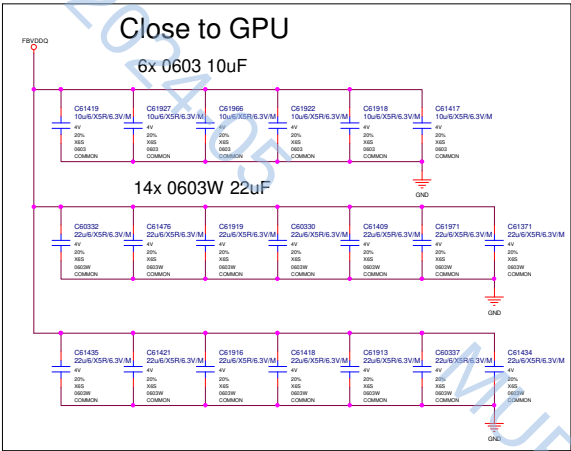


14x 0603 10uF

52 x 0402 1uF



FBVDDQ



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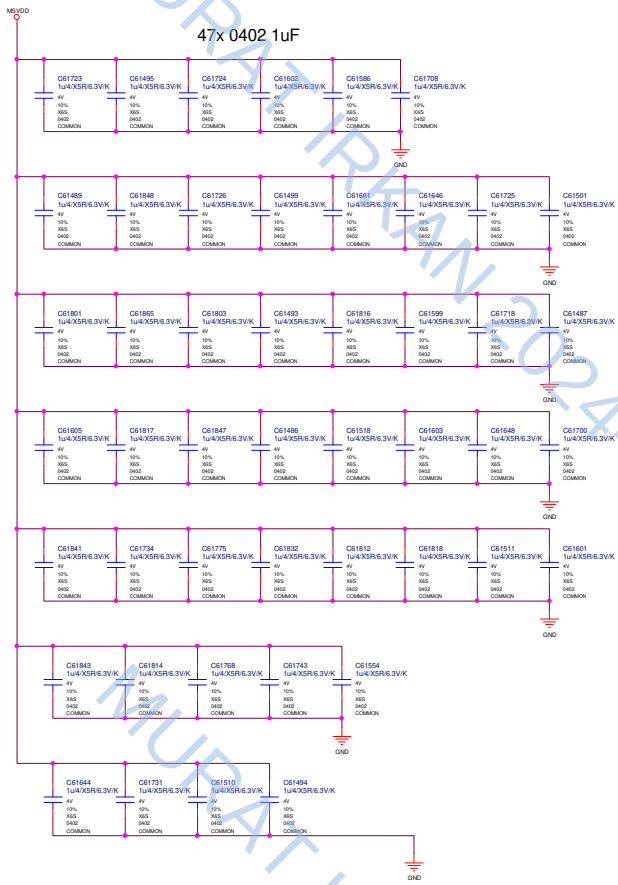
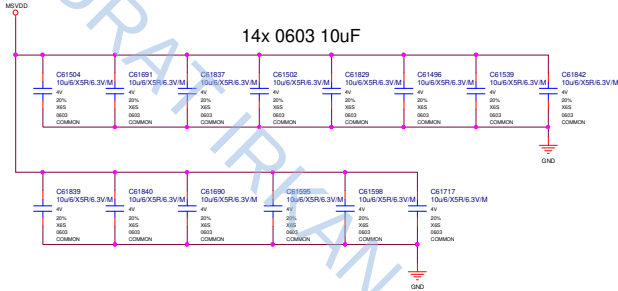
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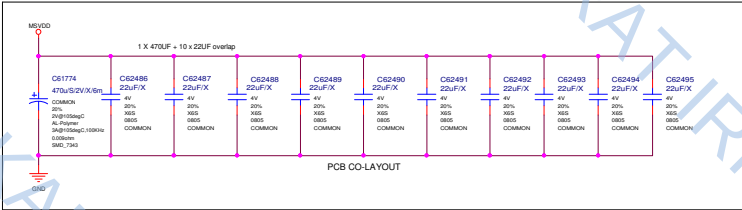
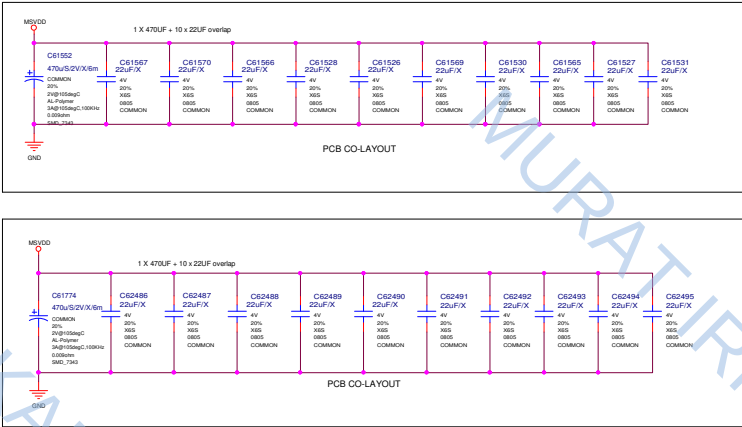
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Part	PCIE		
Docu	Document Number		
Part	GV-N3090EAGLE-24GD		
Date	Wednesday, September 20, 2023		
Rev	1.0		

UNDER GPU

MSVDD



2x SMD7343 220uF (EACH CO-LAYOUT WITH 10x 22uF 0805)



REMOVED 3X 1uF 0402 DUE TO SPACE RESTRICTION

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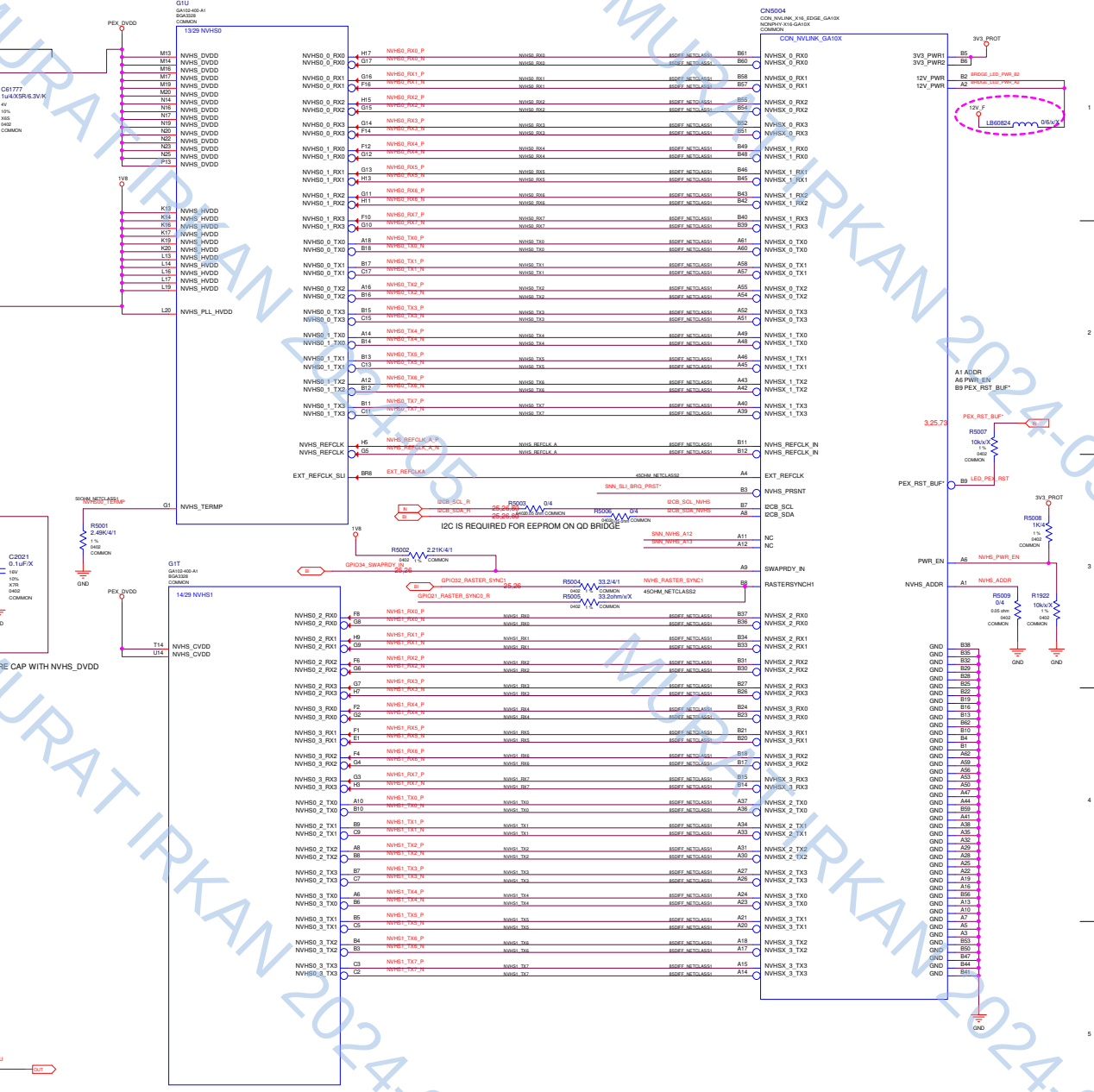
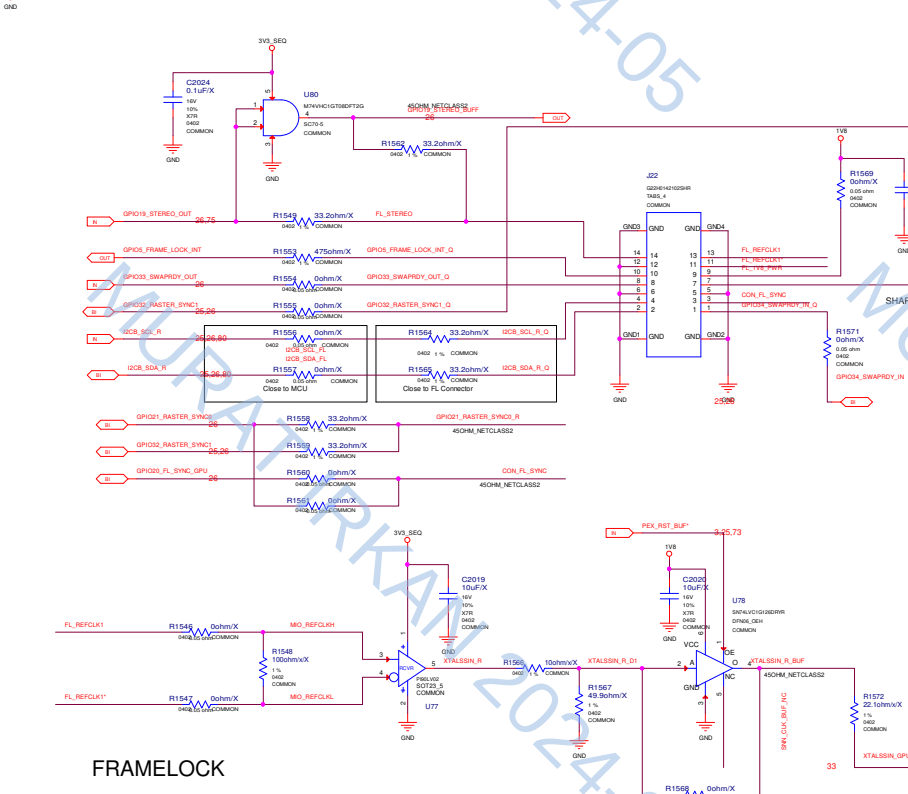
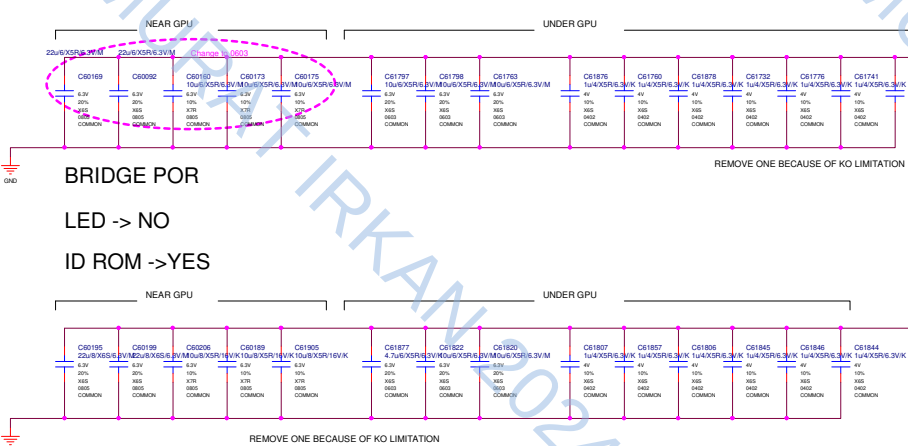
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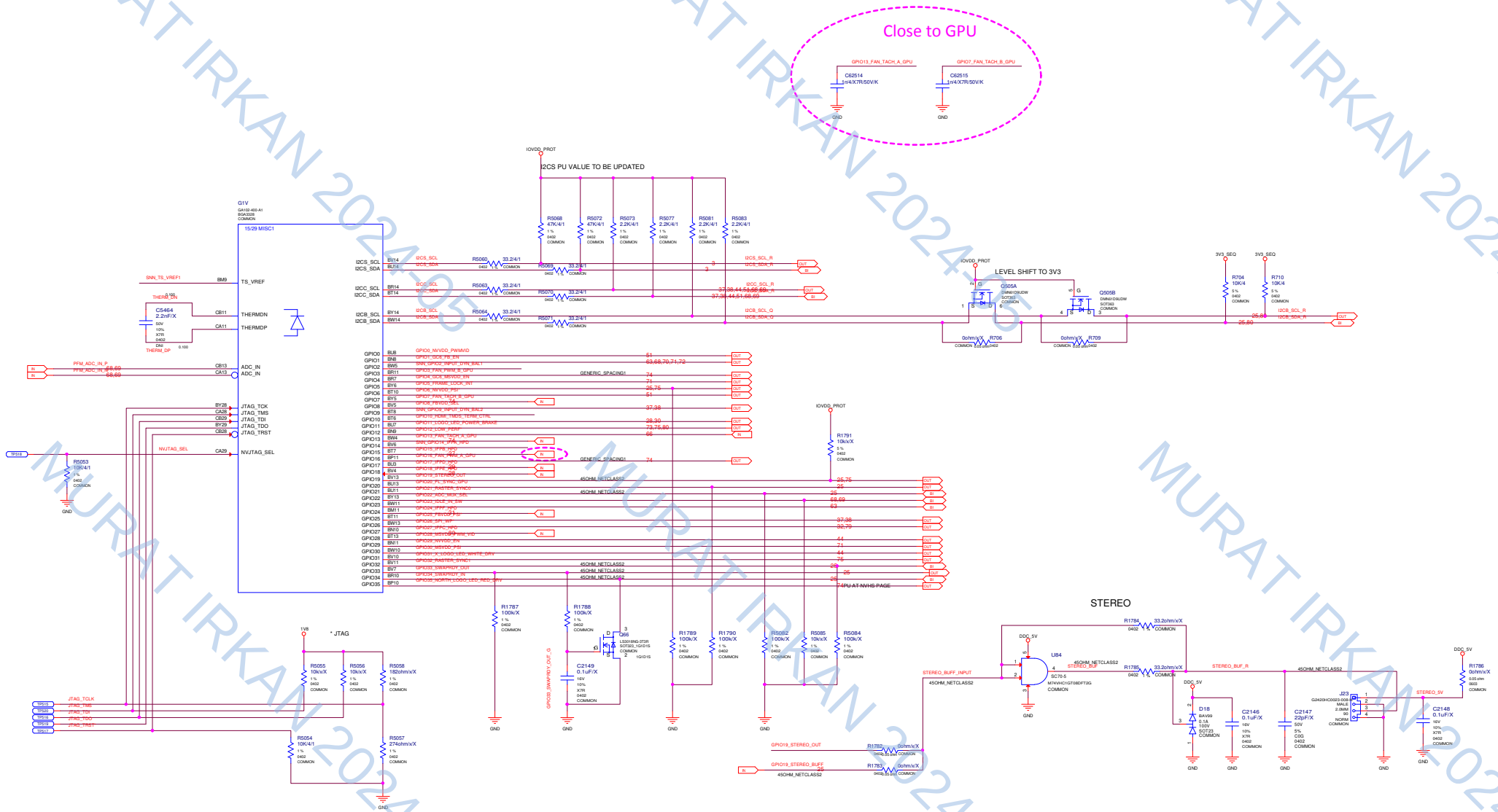
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Title		PCIE	
Size	Document Number	Rev	
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NVHS Interface and FRAME LOCK



MISC1: Fan, Thermal, JTAG, GPIO, STEREO



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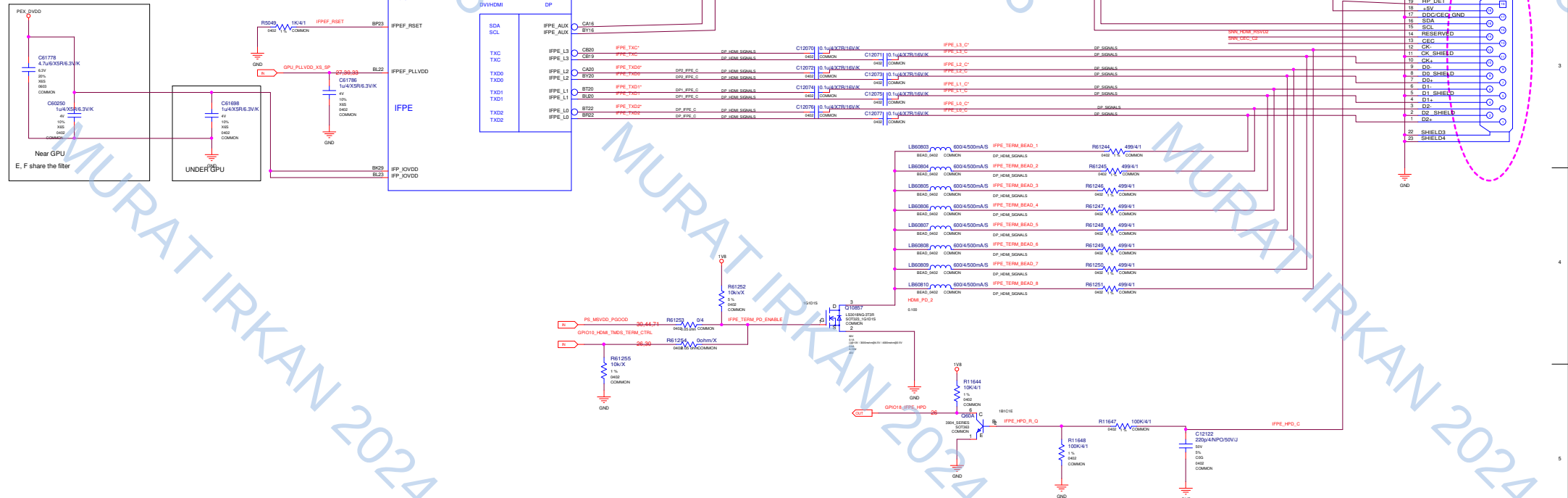
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MISC: THERMAL, JTAG, GPIO

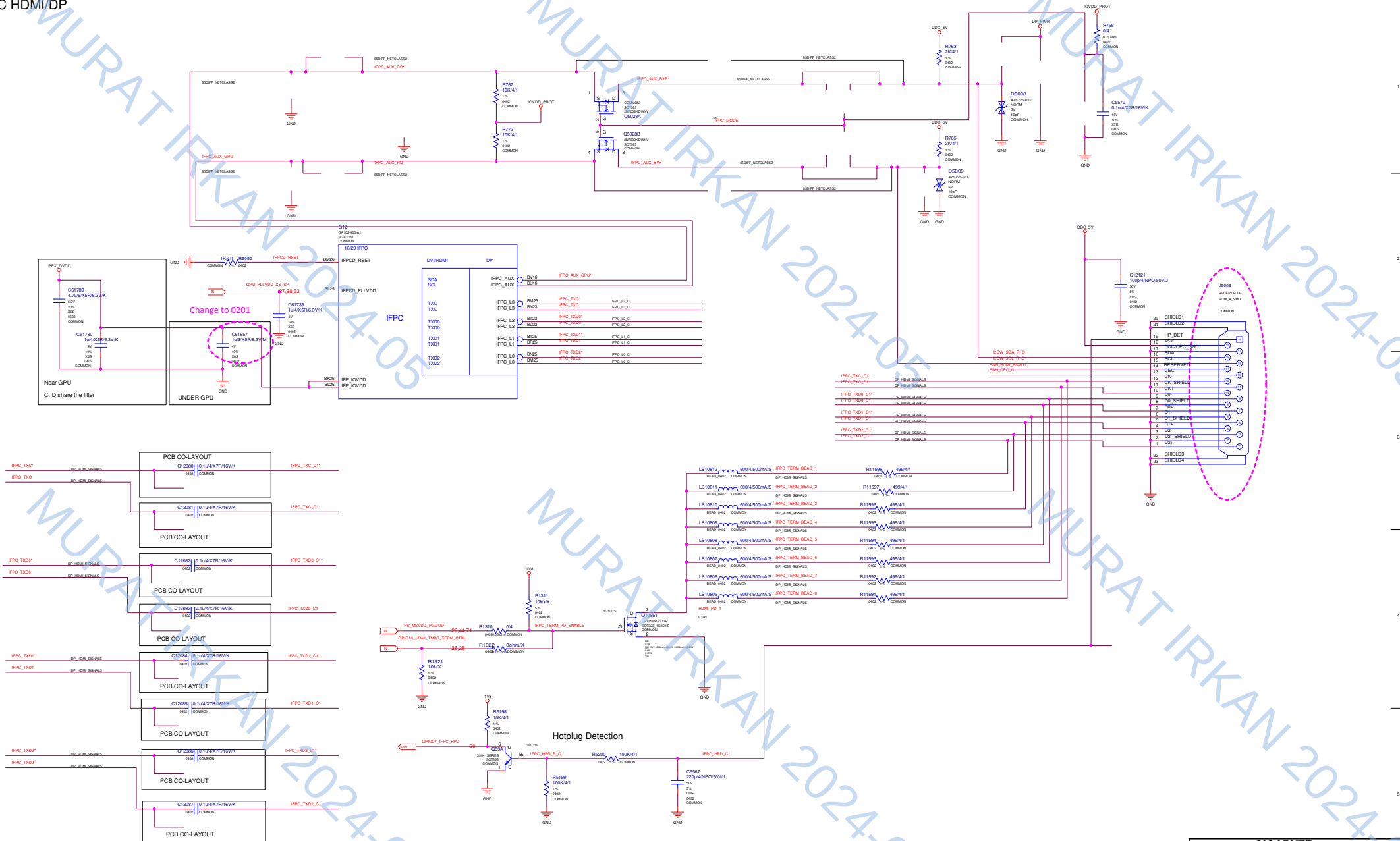
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Size	Document Number	Rev	
Customer	GV-N3090EAGLE-24GD	1.0	
Date:	Wednesday, September 30, 2020	Sheet	26 of 82

Change to HDMI



IFPC HDMI/DP

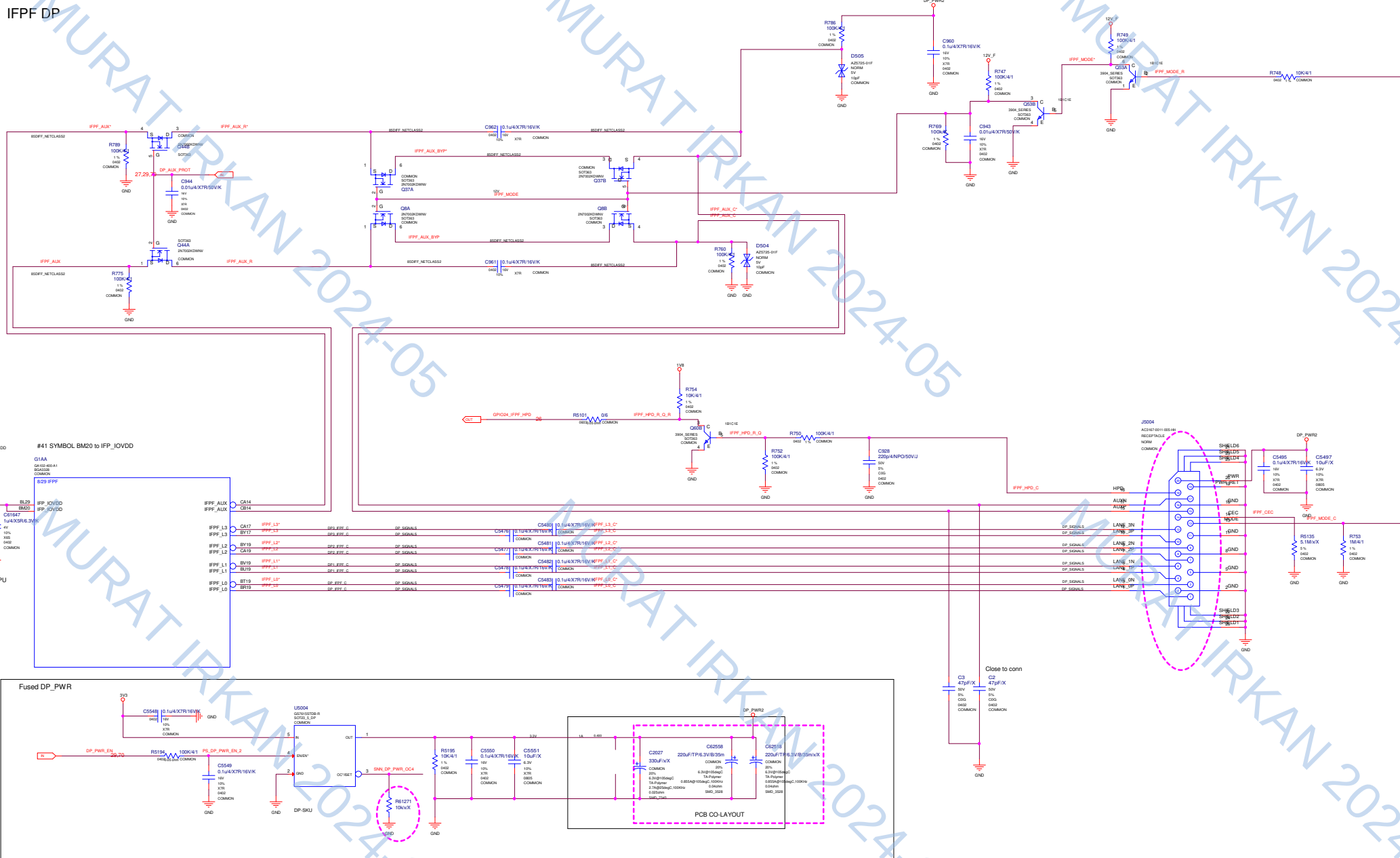


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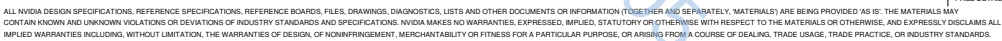
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GIGABYTE



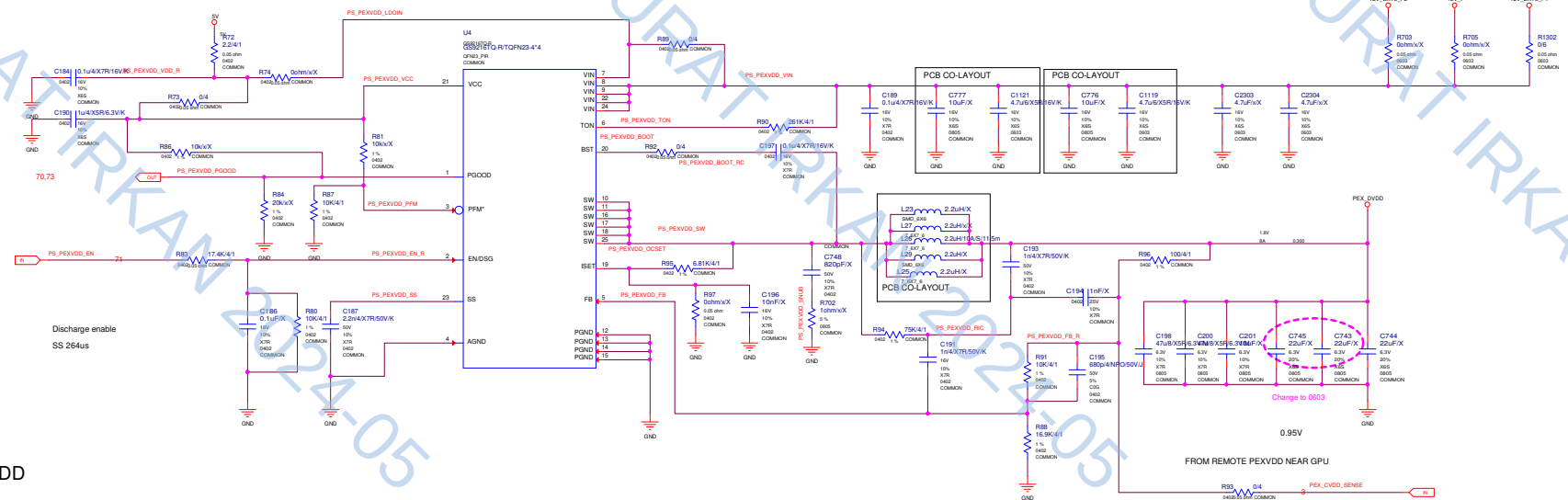
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Size Custom	Document Number GV-N3090EAGLE-24GD		Rev 1.0
Date:	Wednesday, September 30, 2020	Sheet	31 of 82

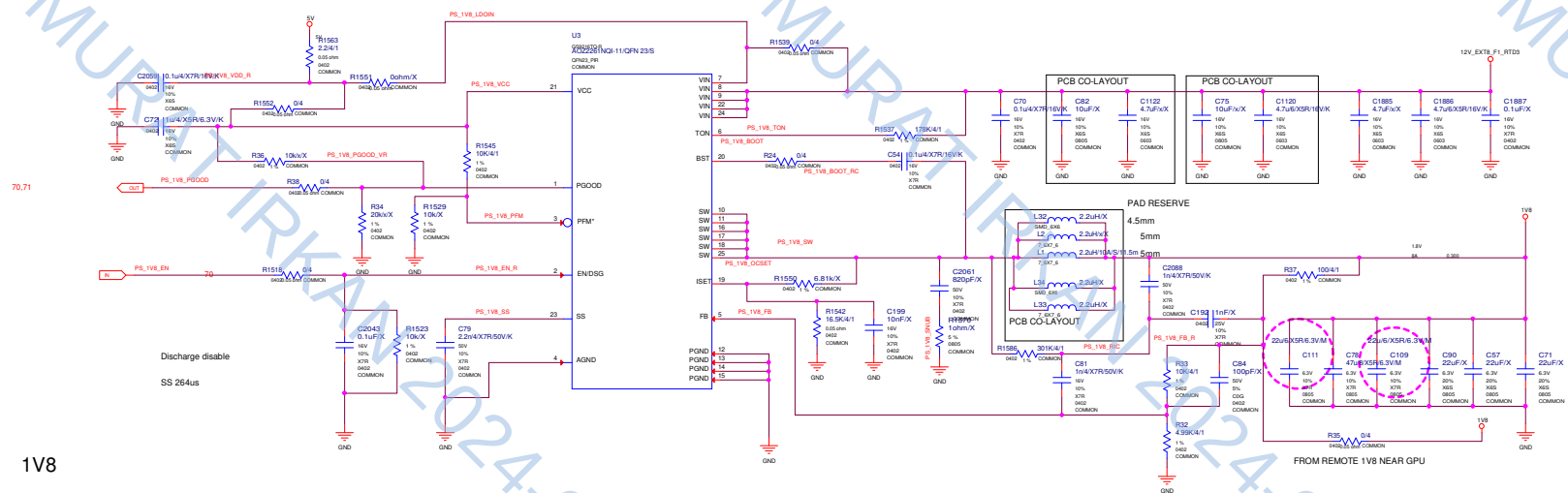
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Size	Document Number		Rev
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Date:	Wednesday, September 30, 2020	Sheet	33 of 82

PS: PEX_DVDD 1V8 Rail



PEX_DVDD



1V8

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Title PCIE			
Size	Document Number		Rev
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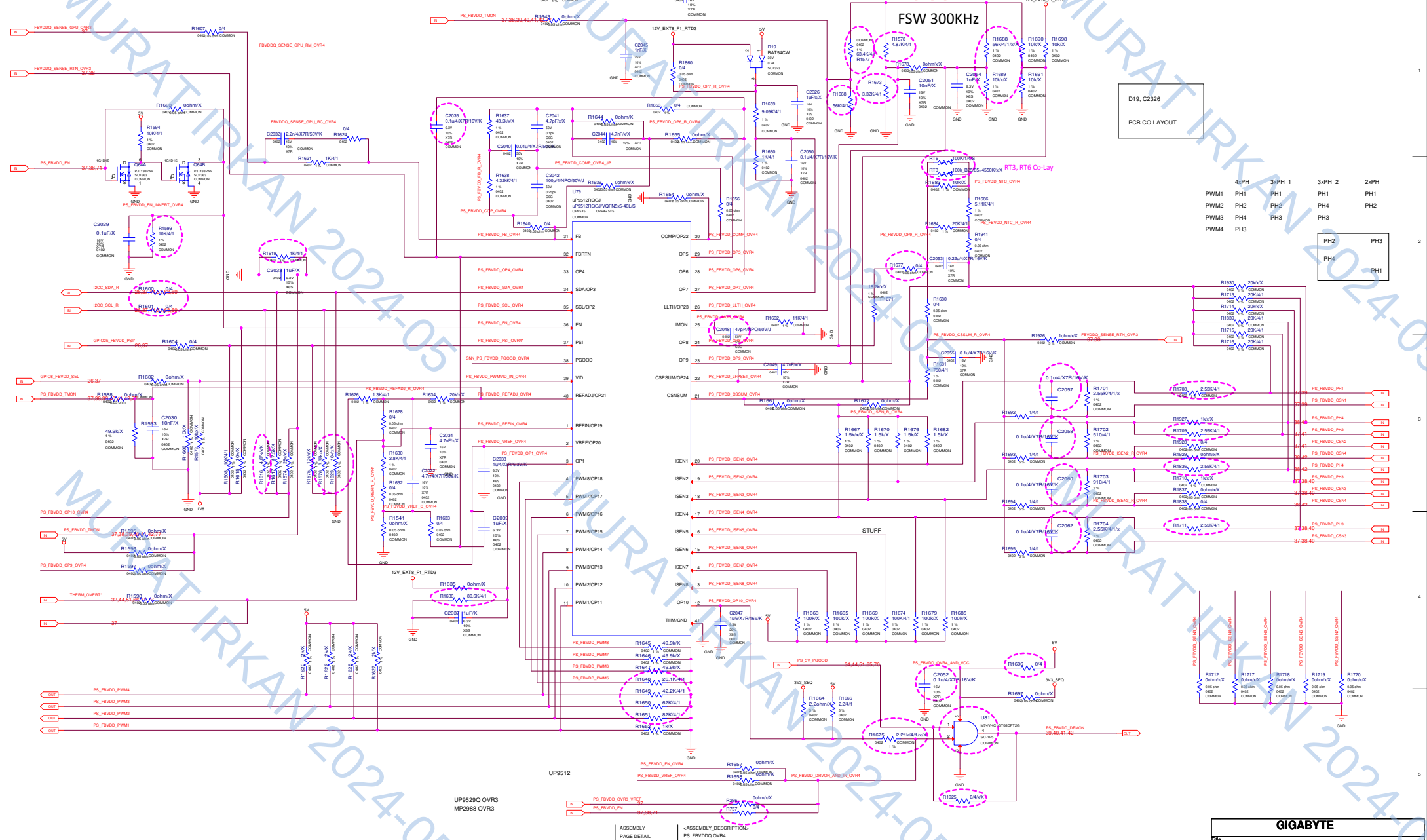
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PCIE			
Size	Document Number		Rev
Custom	GV-N3090EAGLE-24GD		1.0
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PS:FBVDDQ OVR4



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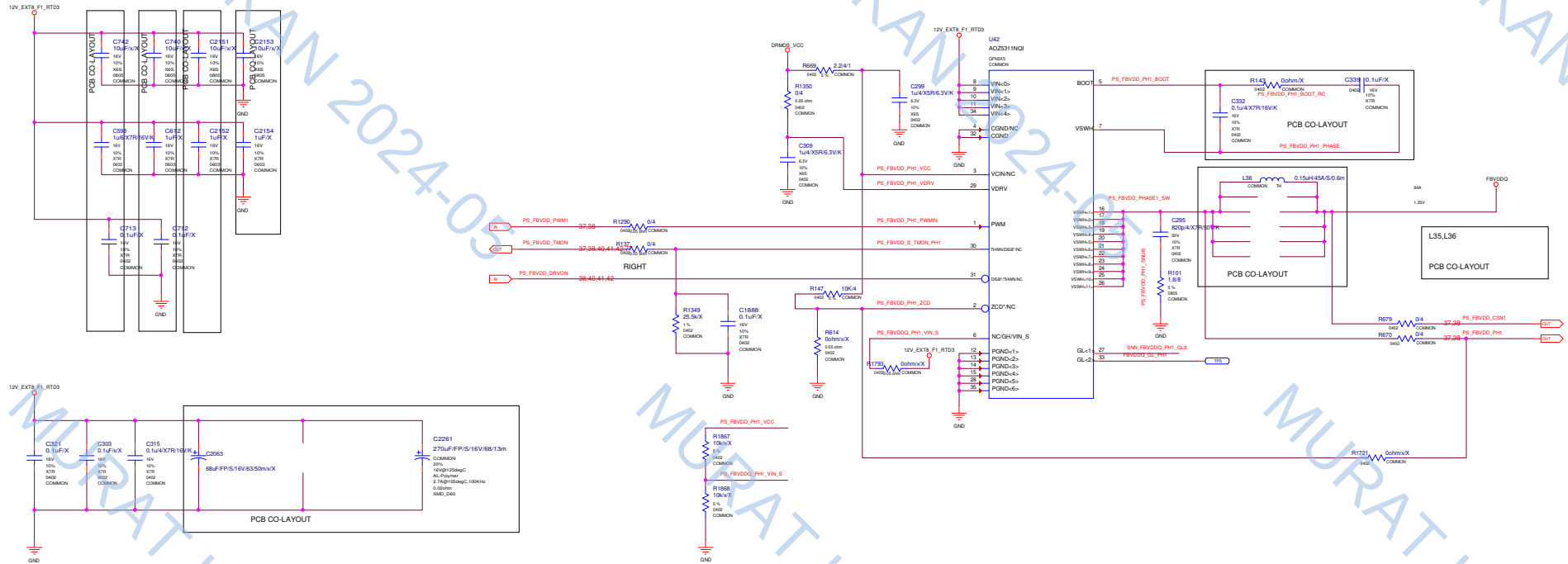
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GIGABYTE

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PCIE			
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		GV-N3090EAGLE-24GD	1.0
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PS: FBVDD PH1

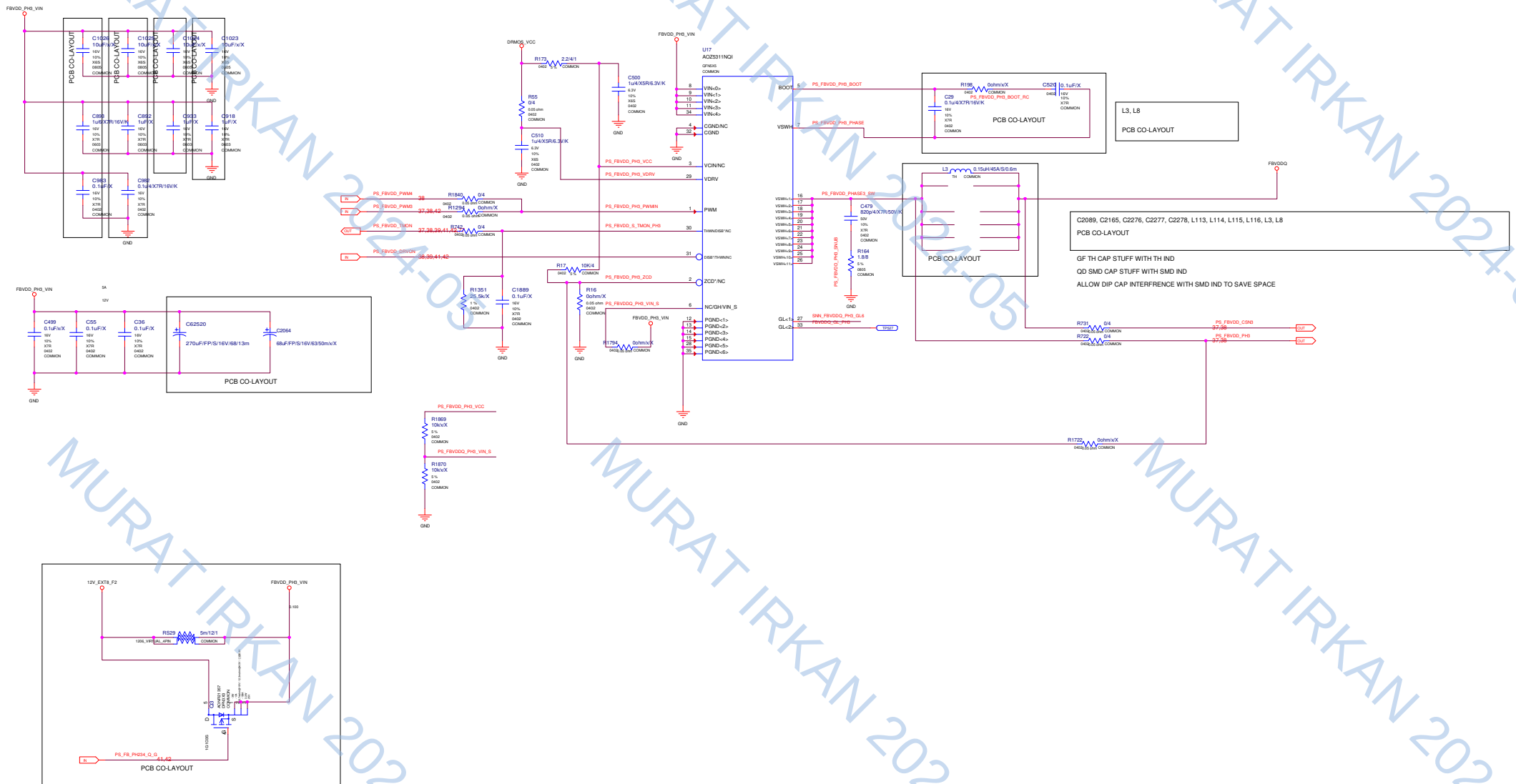


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GIGABYTE			
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Size	Document Number		Rev
Custom	GV-N3090EAGLE-24GD		1.0
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PS: FBVDD PH3



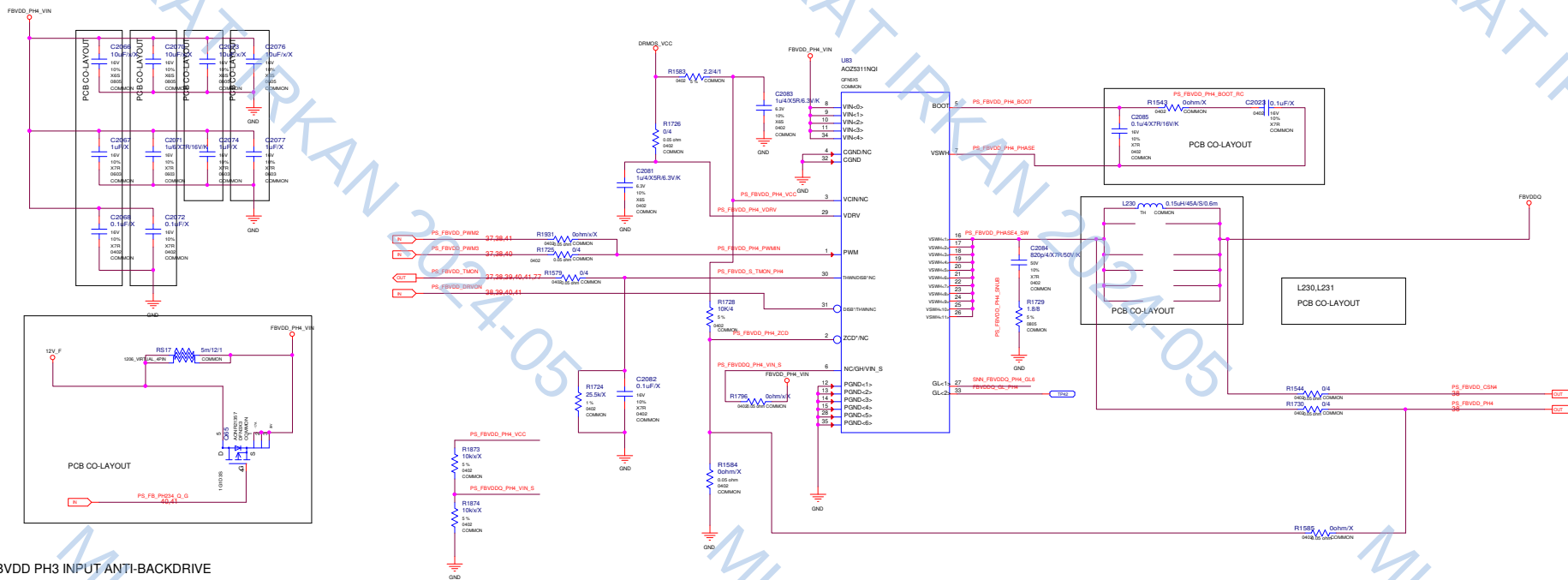
FBVDD PH3 INPUT ANTI-BACKDRIVE

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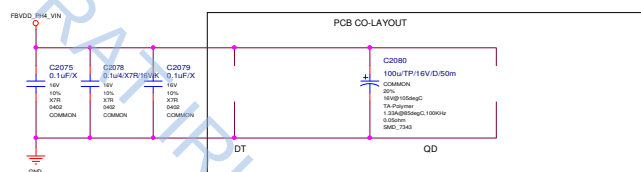
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Size	Document Number		Rev
Custom	GV-N3090EAGLE-24GD		1.0
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PS: FBVDD PH4



FBVDD PH3 INPUT ANTI-BACKDRIVE



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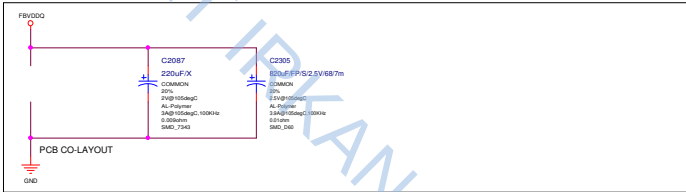
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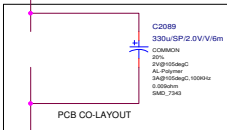
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Custom	GV-N3090EAGLE-24GD		1
Date:	Wednesday, September 30, 2020	Sheet	42 of 82

REFER TO PCB FOR BOM STUFFING

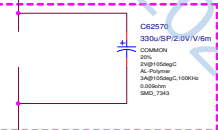
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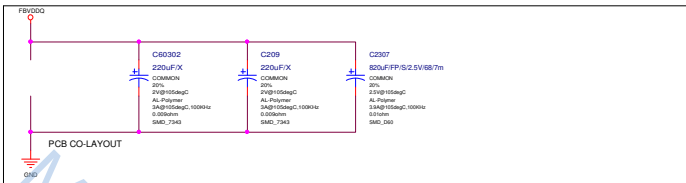
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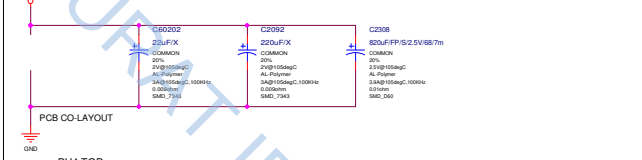
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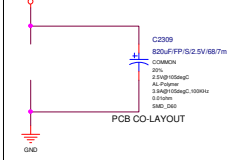
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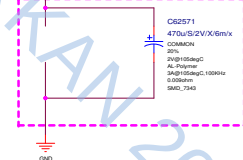
PH4 TOP



PH4 TOP

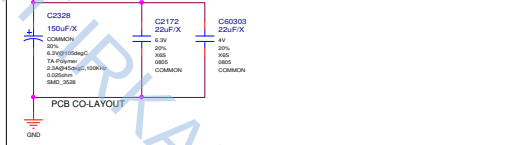


PH4 Bottom

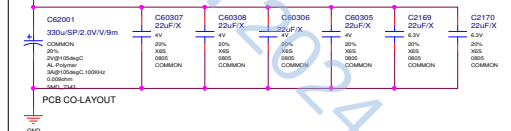


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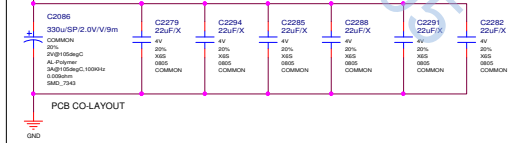
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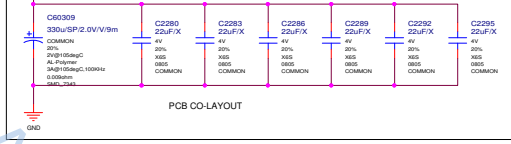
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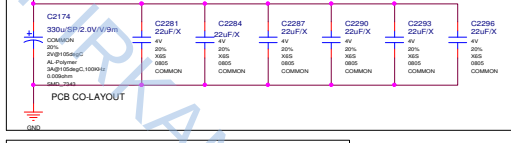
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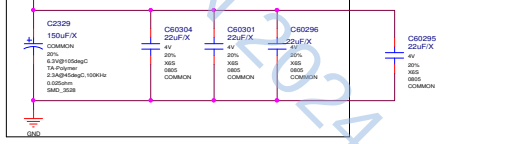
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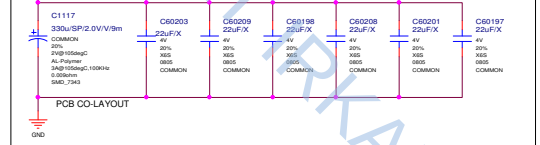
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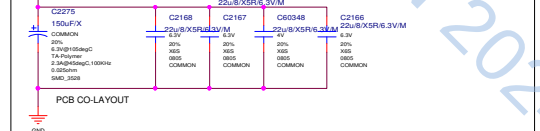
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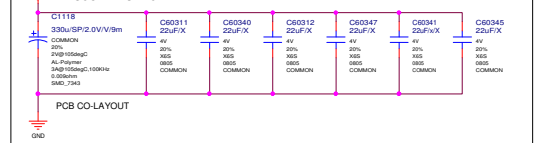
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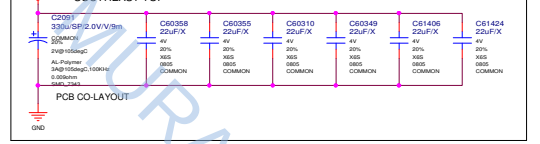
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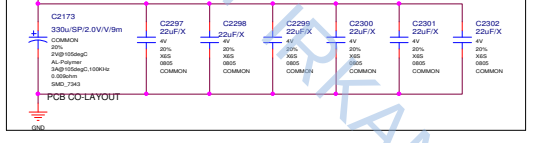
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SOUTHEAST TOP



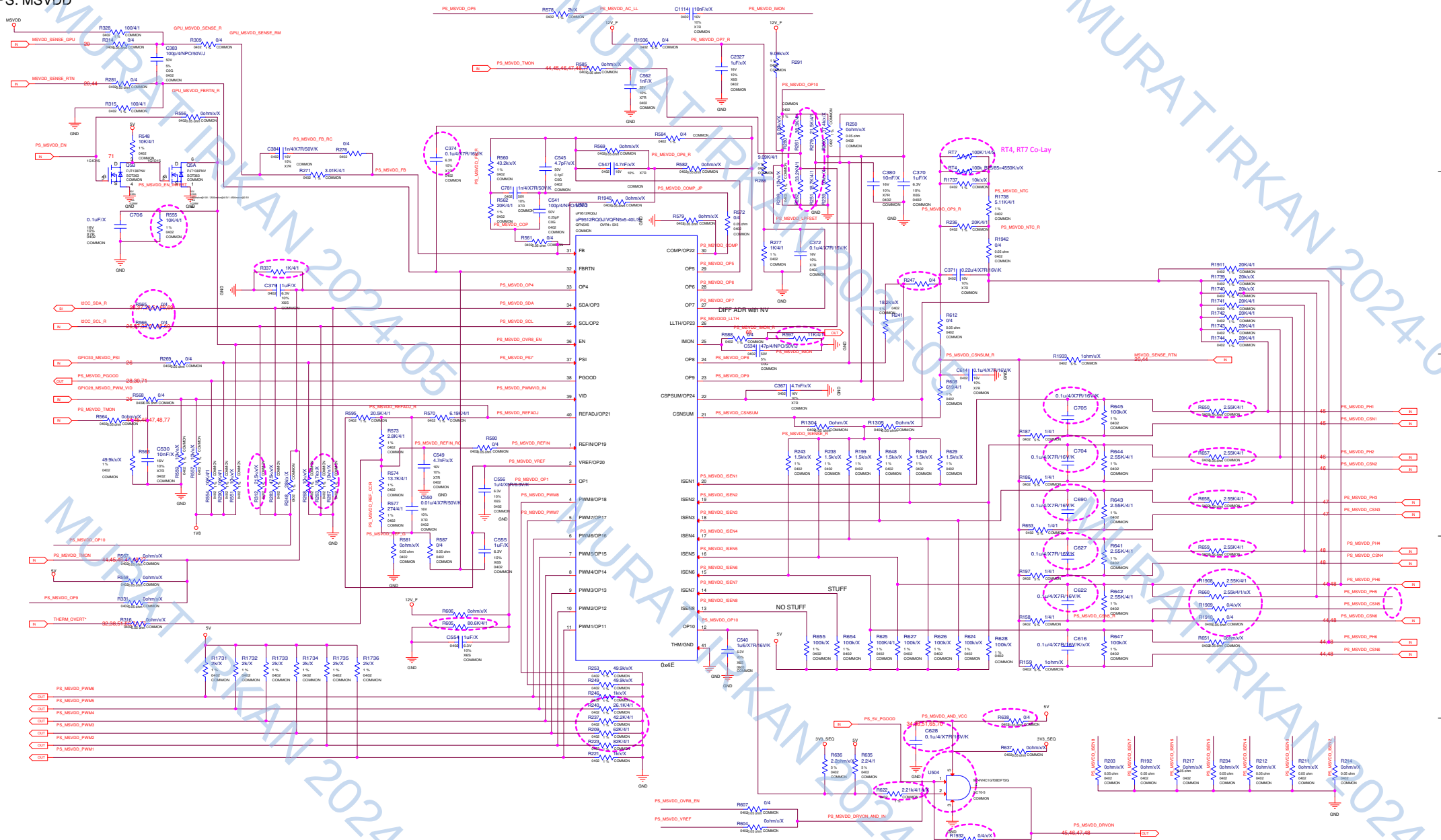
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PCB CO-LAYOUT



PS: MSVDD



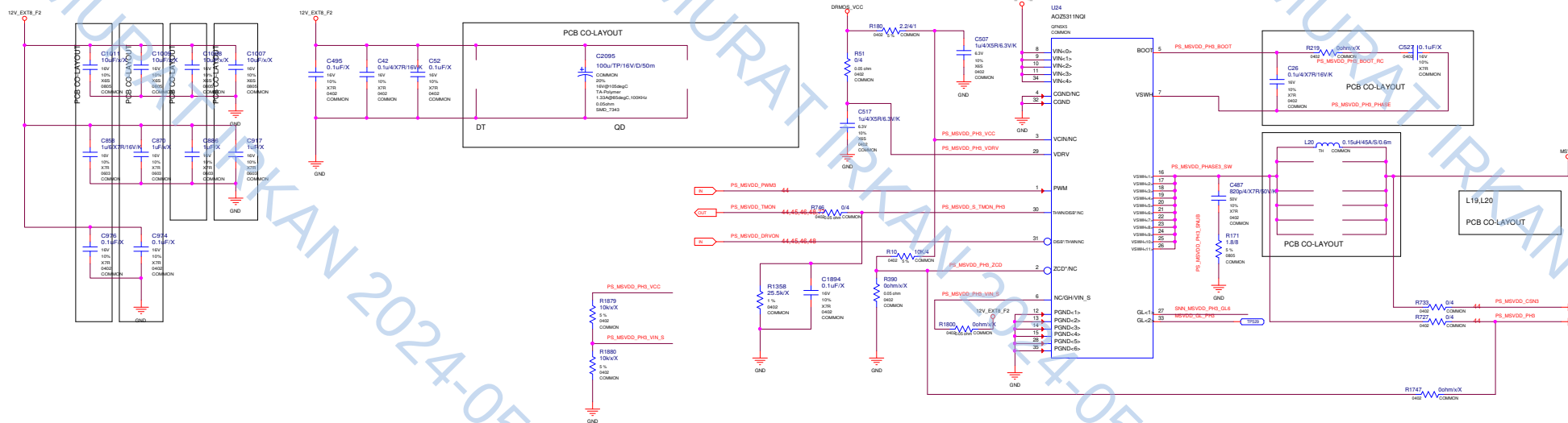
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GIGABYTE

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Size	Document Number						Rev
Custom	GV-N3090EAGLE-24GD						1.
Date:	Wednesday, September 30, 2020		Sheet	44	of	82	

PS:MSVDD Phase 3, 5



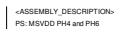
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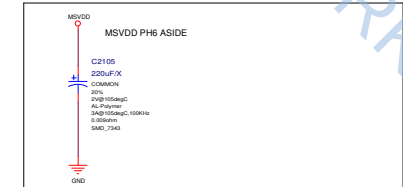
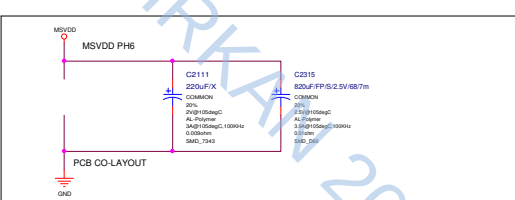
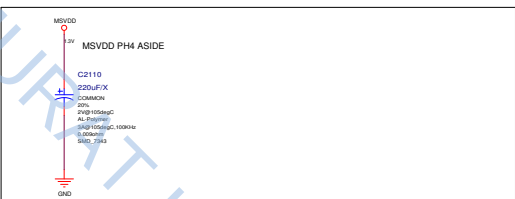
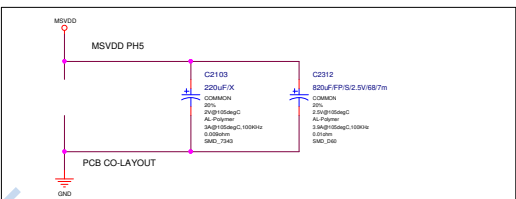
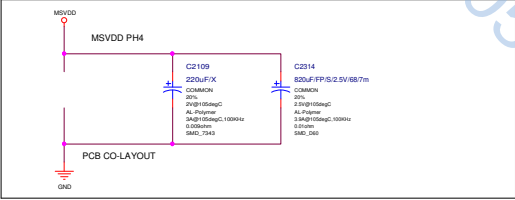
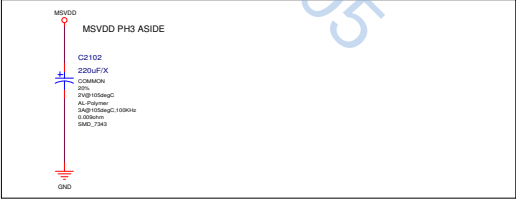
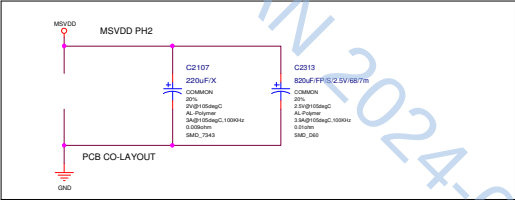
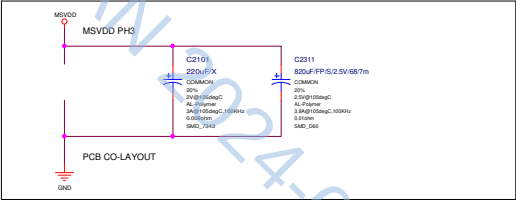
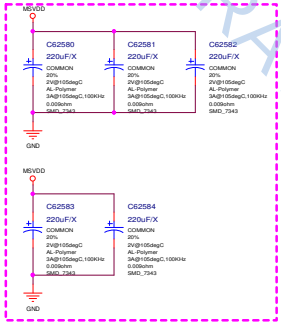
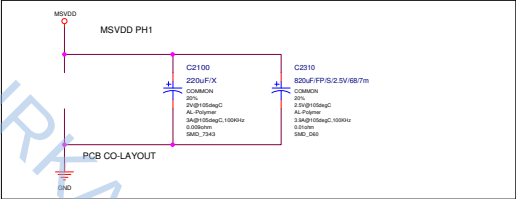
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PS: MSVDD PH3 and PH5

GIGABYTE

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Size	Document Number	Rev	
Custom	GV-N3090EAGLE-24GD	1	
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REFER TO PCB FOR BOM STUFFING



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ASSEMBLY
PAGE DETAIL

ASSEMBLY DESCRIPTION
PS: MSVDD OUTPUT CAP(TOP)

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File	PCIE		
Size	Document Number		
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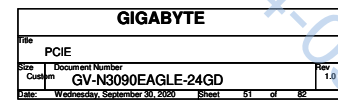
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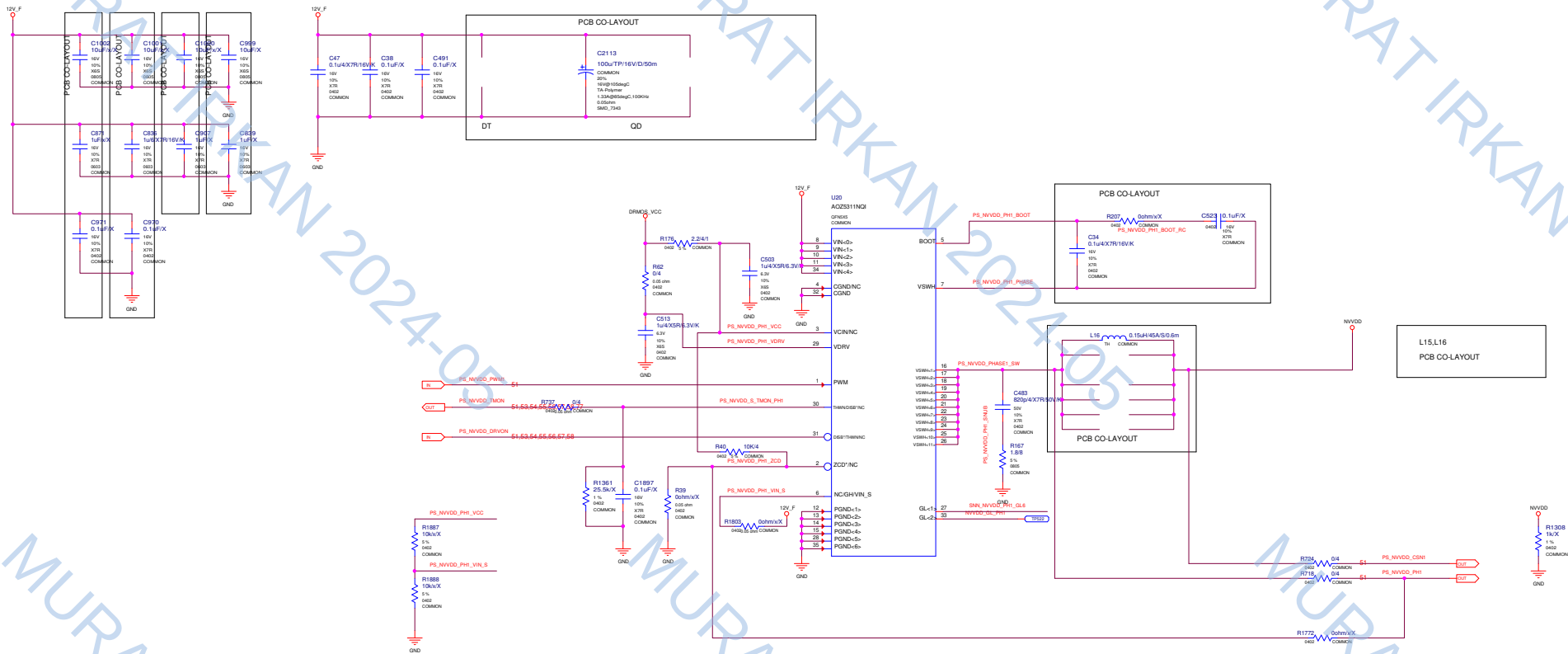
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PS: NVVDD Phase 1



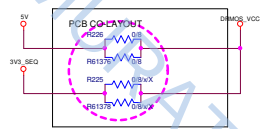
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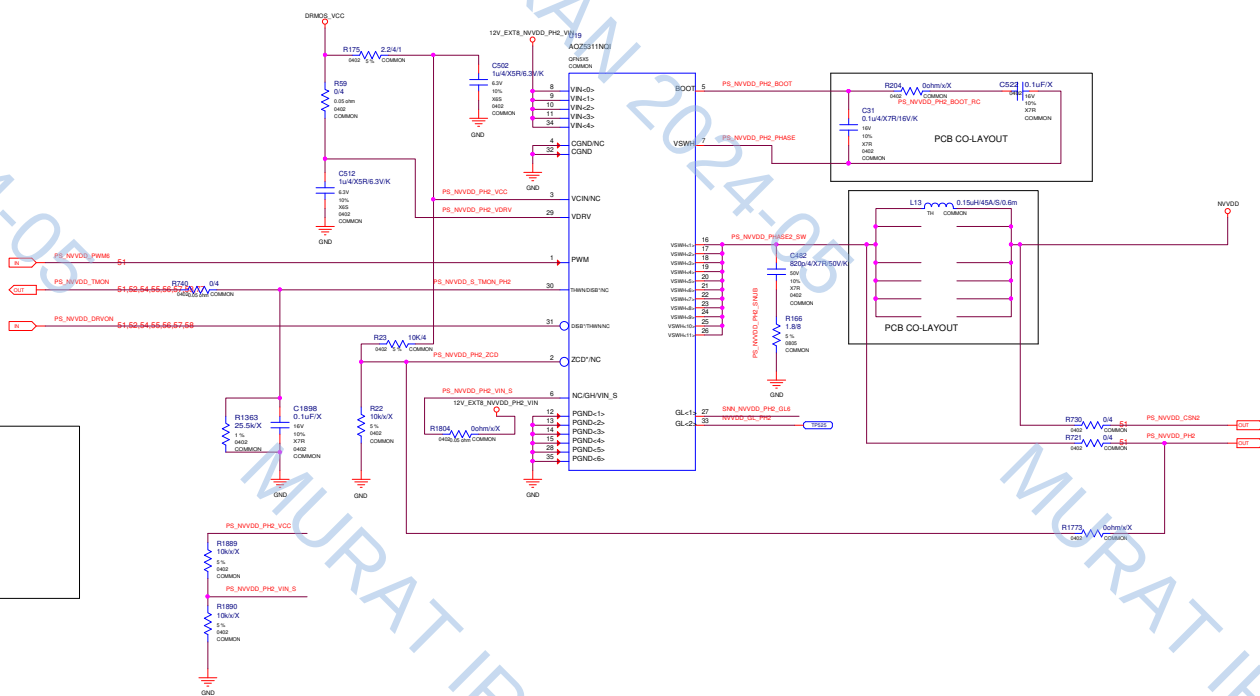
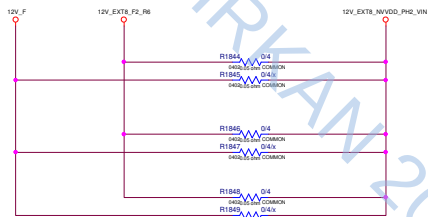
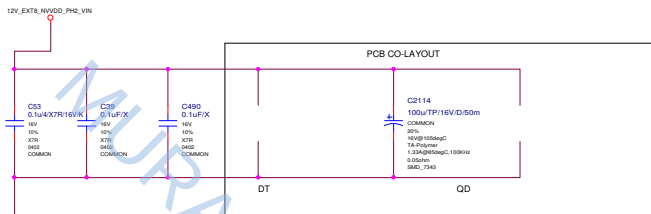
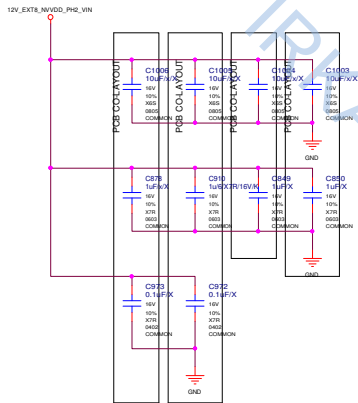
ASSEMBLY DESCRIPTION
PS: NVVDD PH1 (PWMT1)

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PS: NVVDD Phase 2 (PWM6)



MPS DRMOS WITH 3V3 SUPPLY



L13,L14
PCB CO-LAYOUT

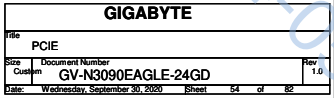
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ASSEMBLY
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PS: NVVDD PH2 (PWM6)

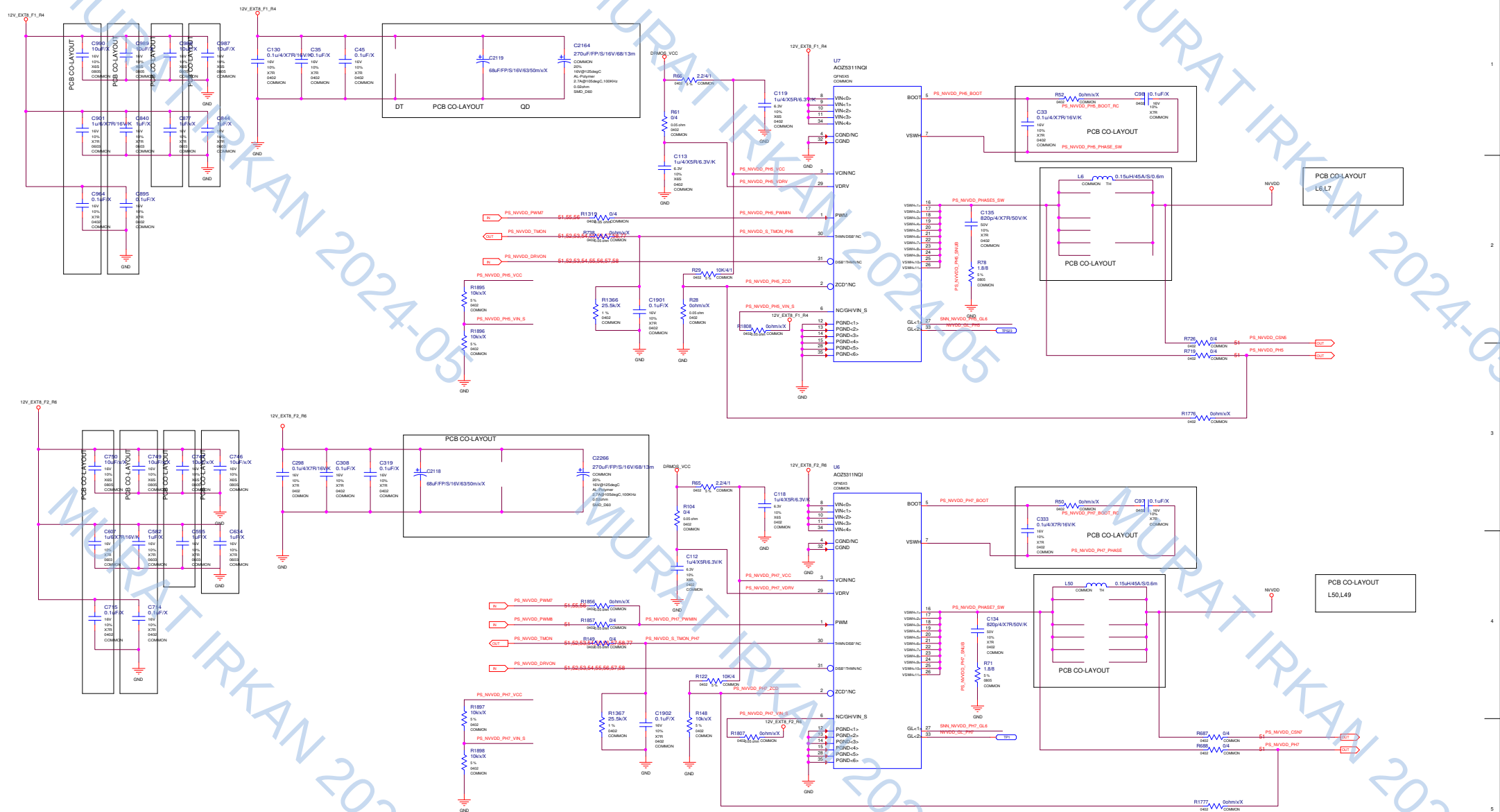
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PS: NVVDD PH 5(PWM4), PH7(PWM8/4)



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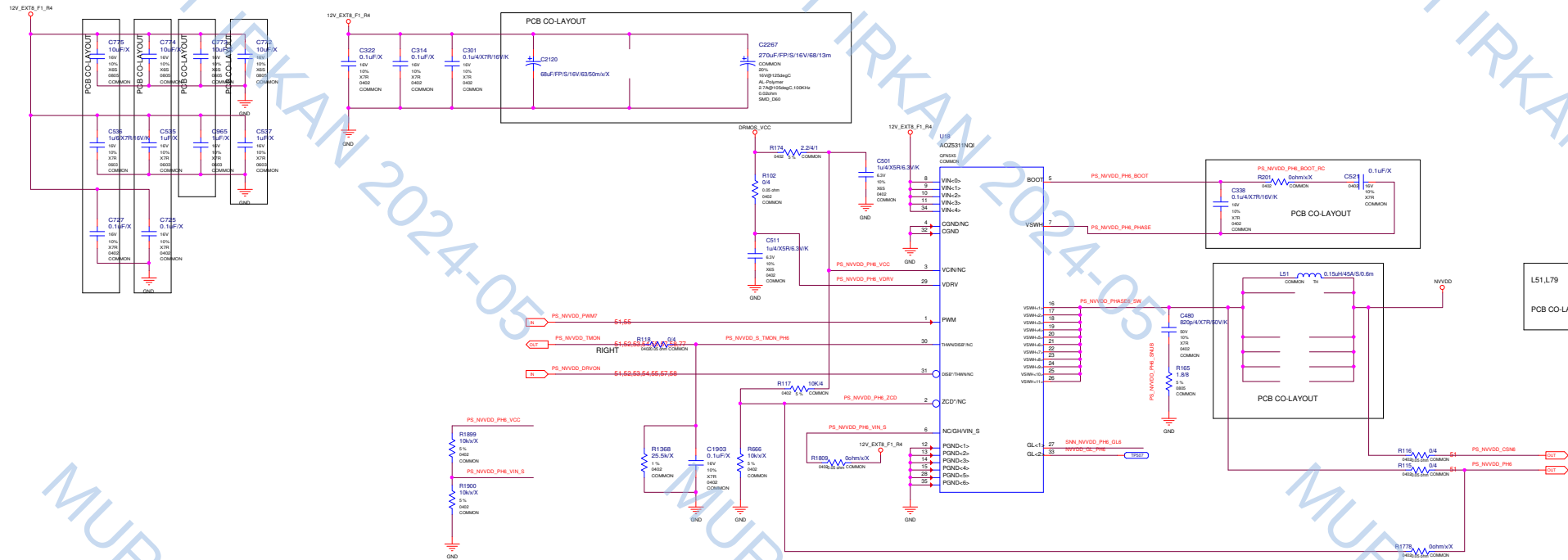
ASSEMBLY
PAGE DETAIL

PS: NVVDD PH5 (PWM7) and PH7 (PWM8)

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PCIE			
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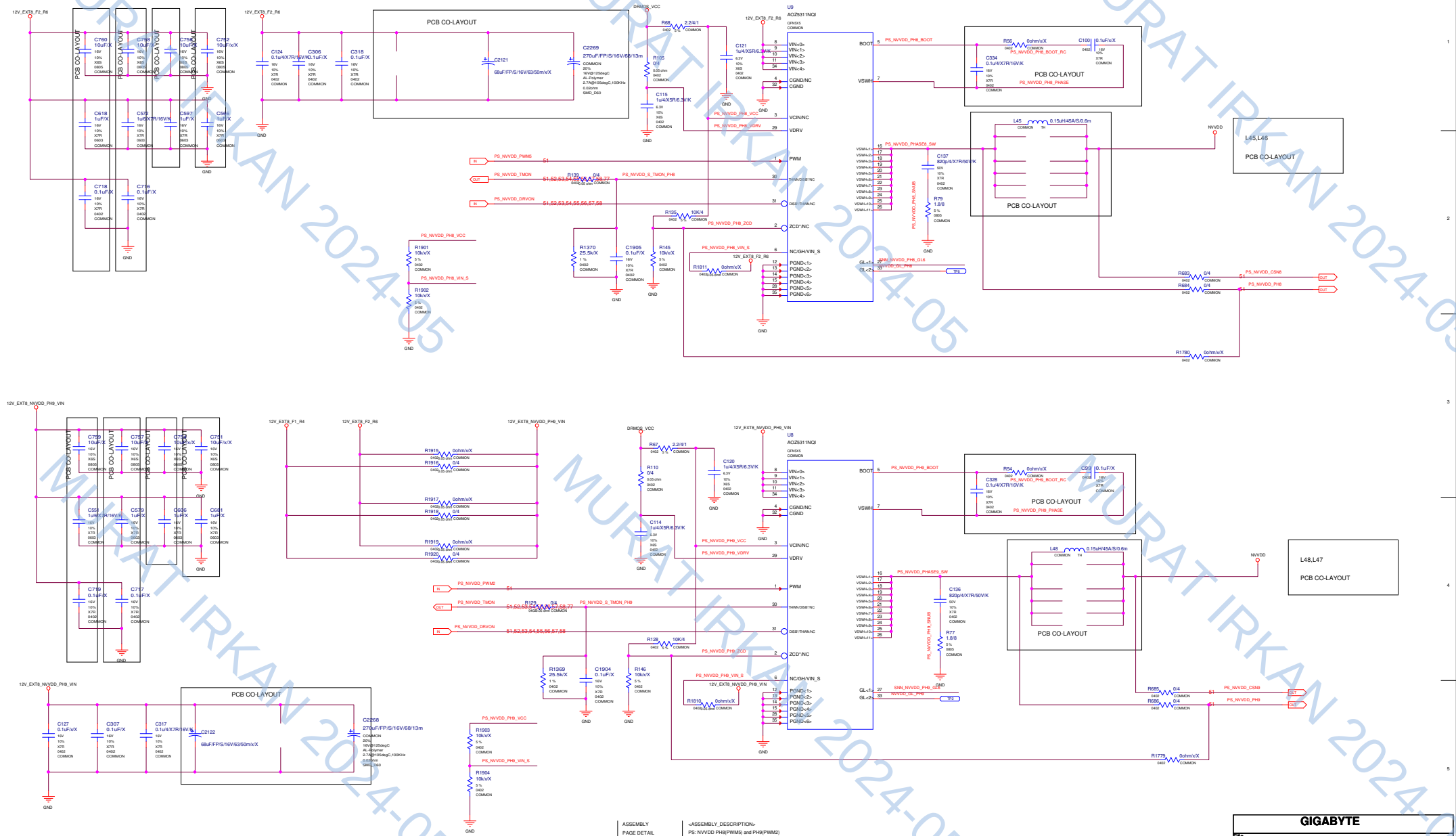
PS: NVVDD Phase 6(PWM4)



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PS: NVVDD PH6 (PWM7)

PS: NVVDD PH 8 (PWM5), PH 9(PWM2)

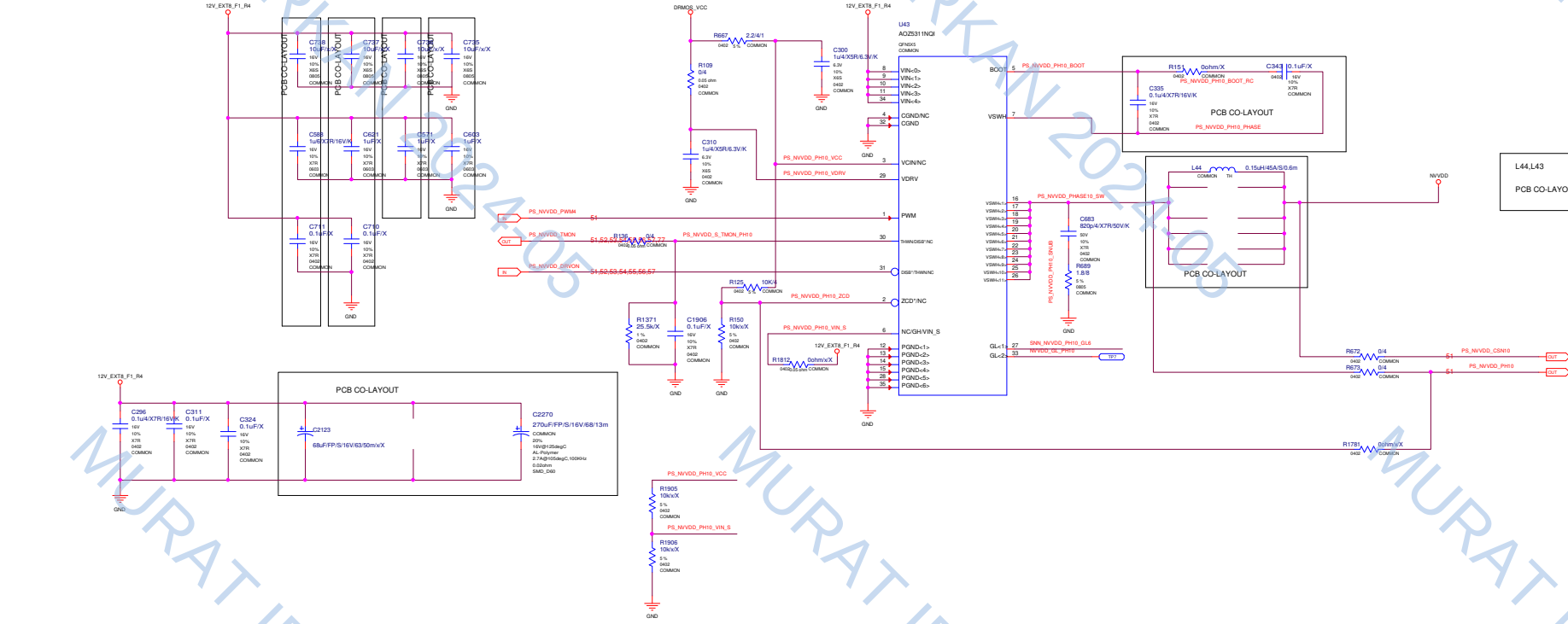


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PS: NVVDD PH8(PWM5) and PH9(PWM2)

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ASSEMBLY
PAGE DETAIL
PS: NVVDD PH10 (PWM7)

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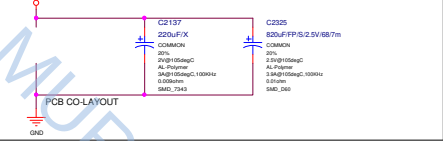
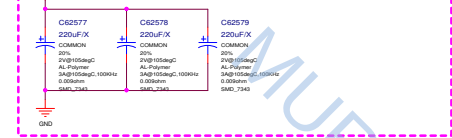
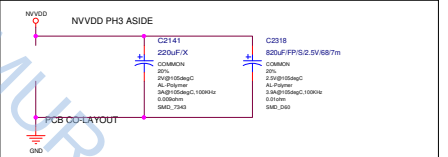
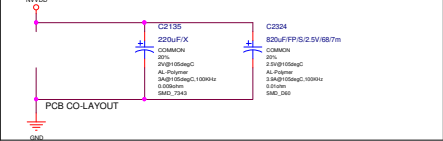
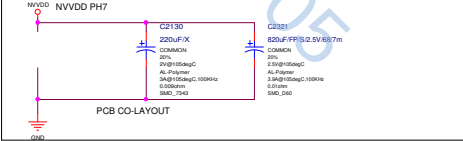
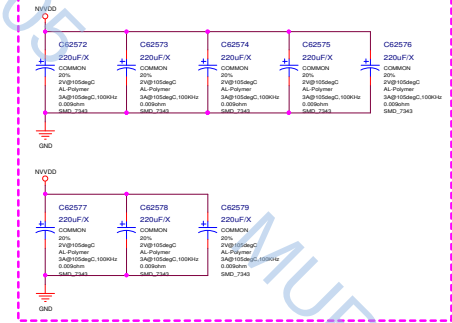
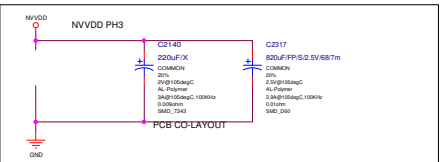
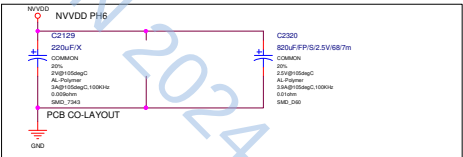
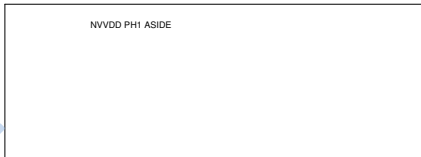
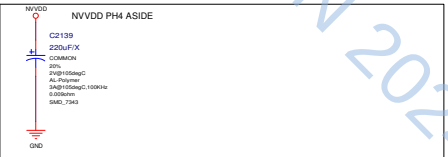
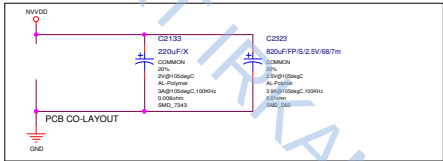
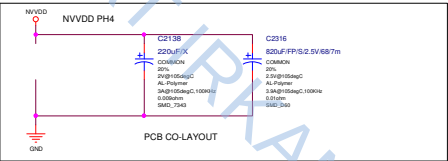
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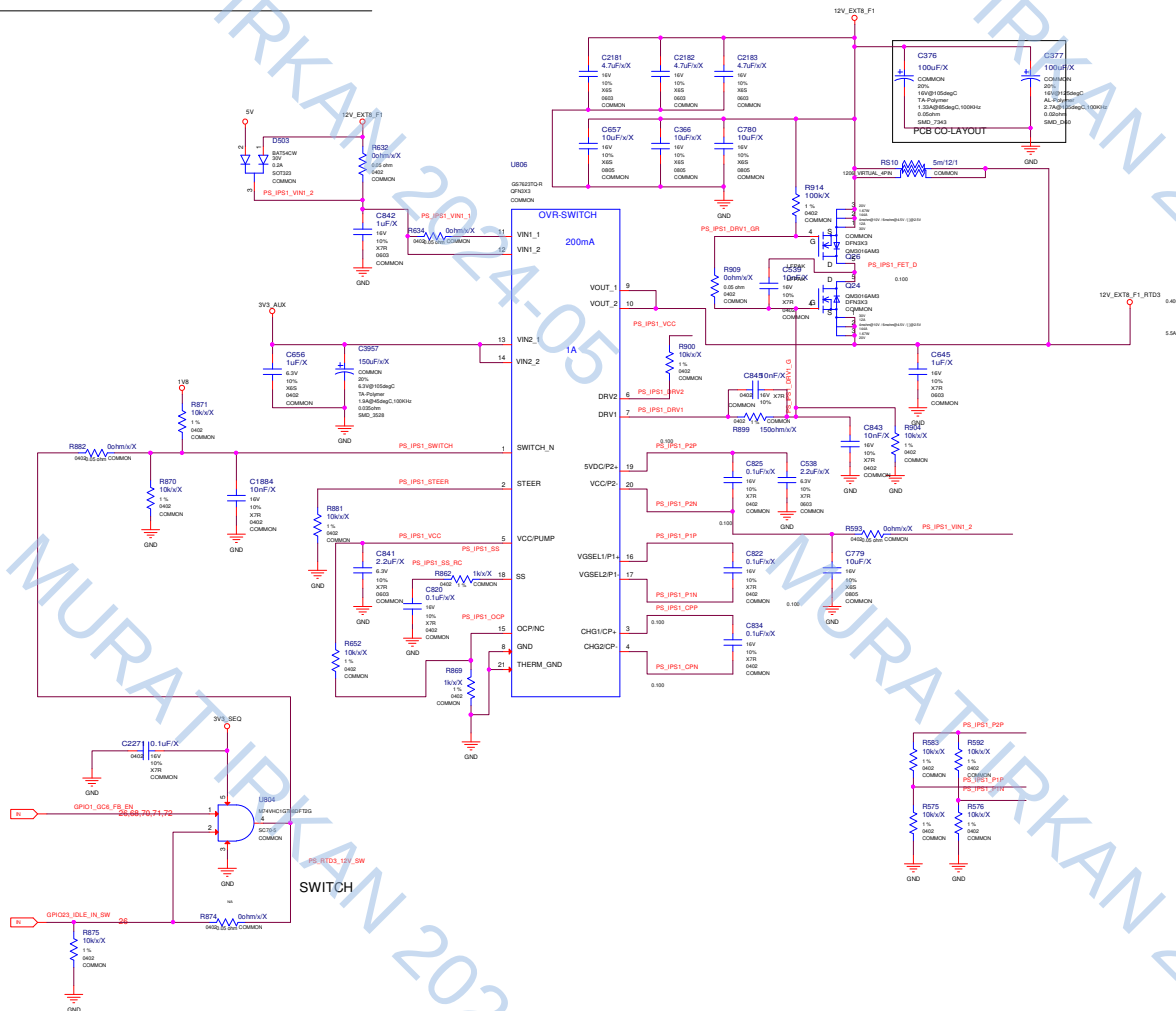
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PS: INPUT SWITCH RTD3

AND GATE LOGIC FOR P-BOARD

GPI01	GPI023	SWITCH	VOUT
0	0	0	12V_F
0	1	0	12V_F
1	0	0	12V_F
1	1	1	3V3A



ASSEMBLY	<ASSEMBLY_DESCRIPTION>
PAGE DETAIL	PS: INPUT SWITCH RTD3

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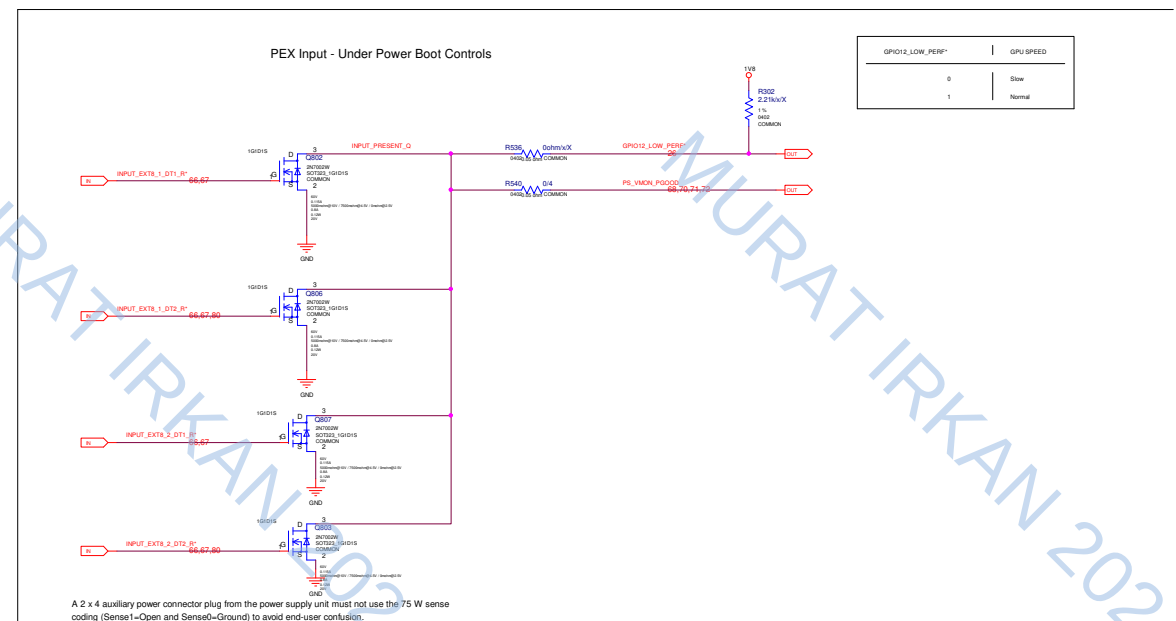
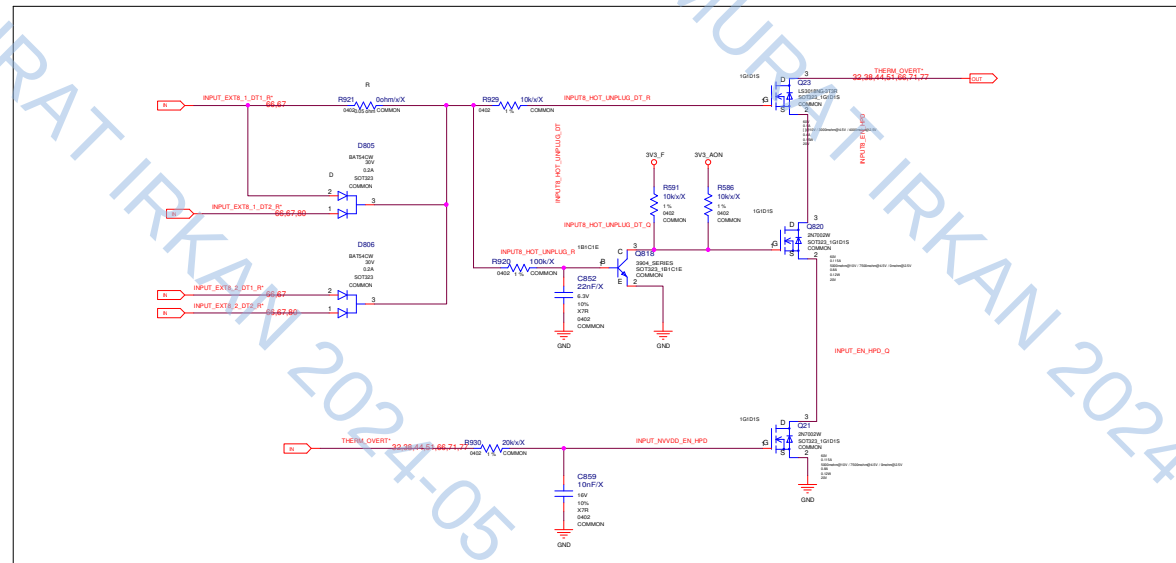
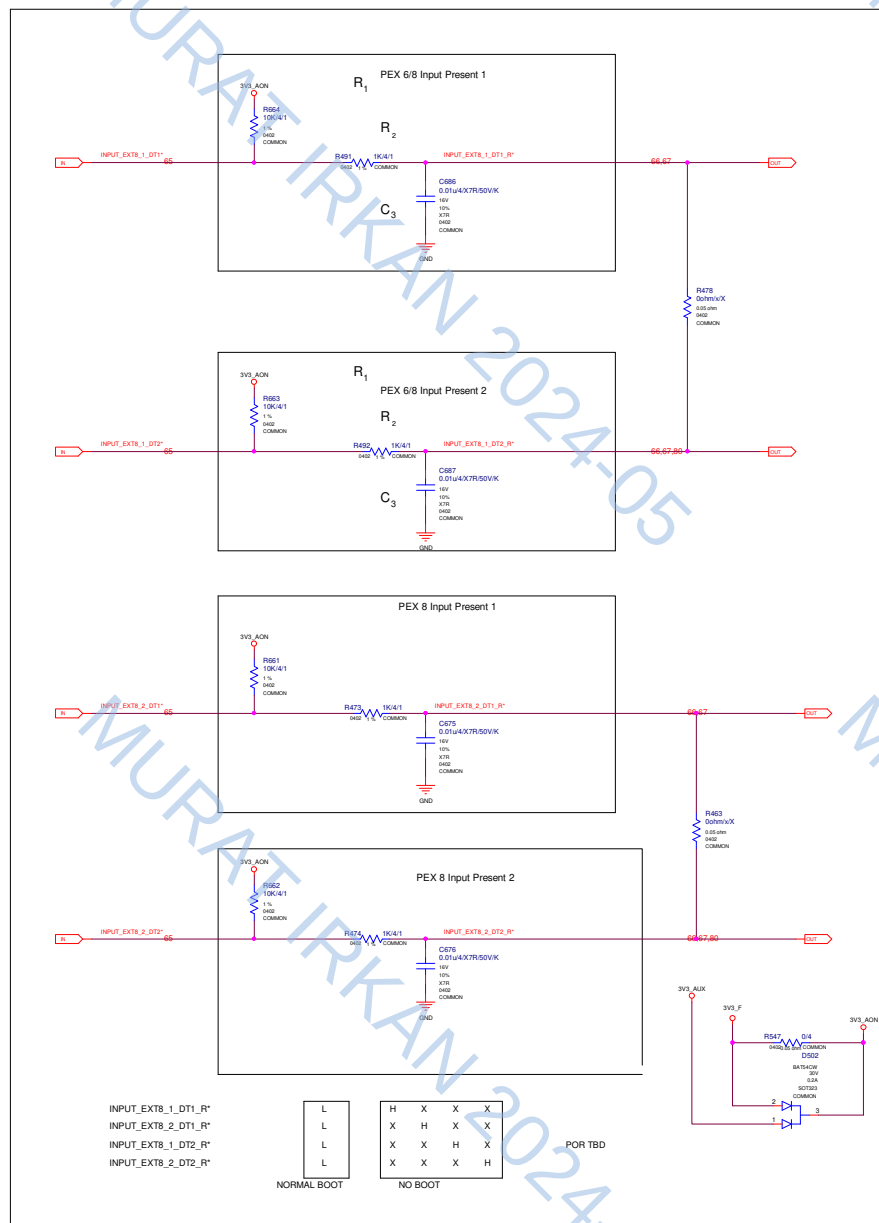
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PS: 12V Current Steering & Hot Unplug Detect



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PS: HOT UNPLUG
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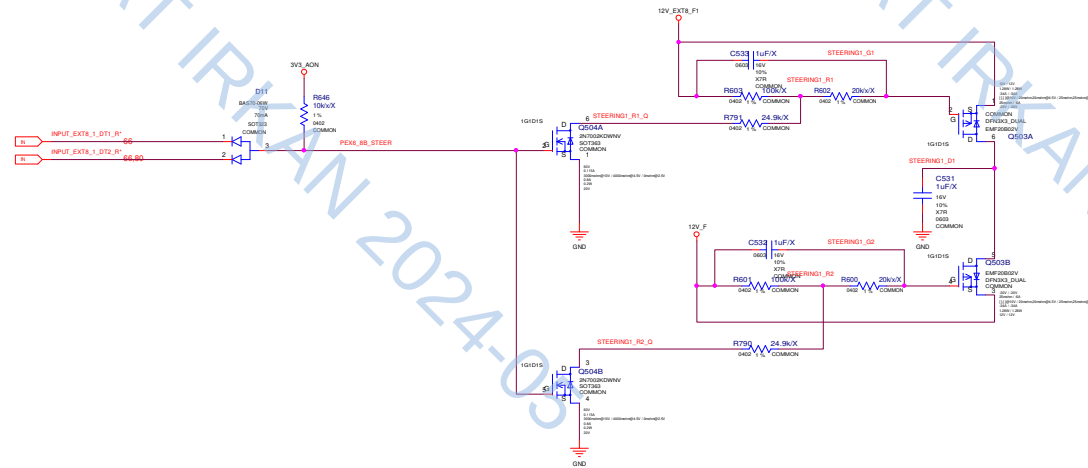
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PS: Discrete Steering

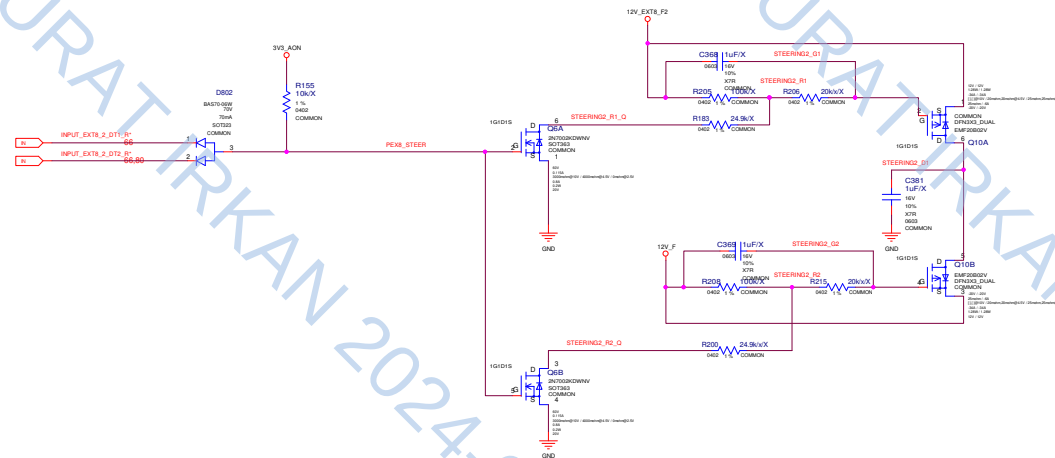
12V CURRENT STEERING (UNDER POWER BOOT):

PEX12V AND 12V_EXT8_F1



NO STUFF BY DEFAULT REF ONLY

PEX12V AND 12V_EXT8_F2



ASSEMBLY
PAGE DETAIL

PS: Discrete Power Steering

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PS: Pre-Filter

STUFF IF USING 2XMUX

Route the trace to PEX Golden Finger

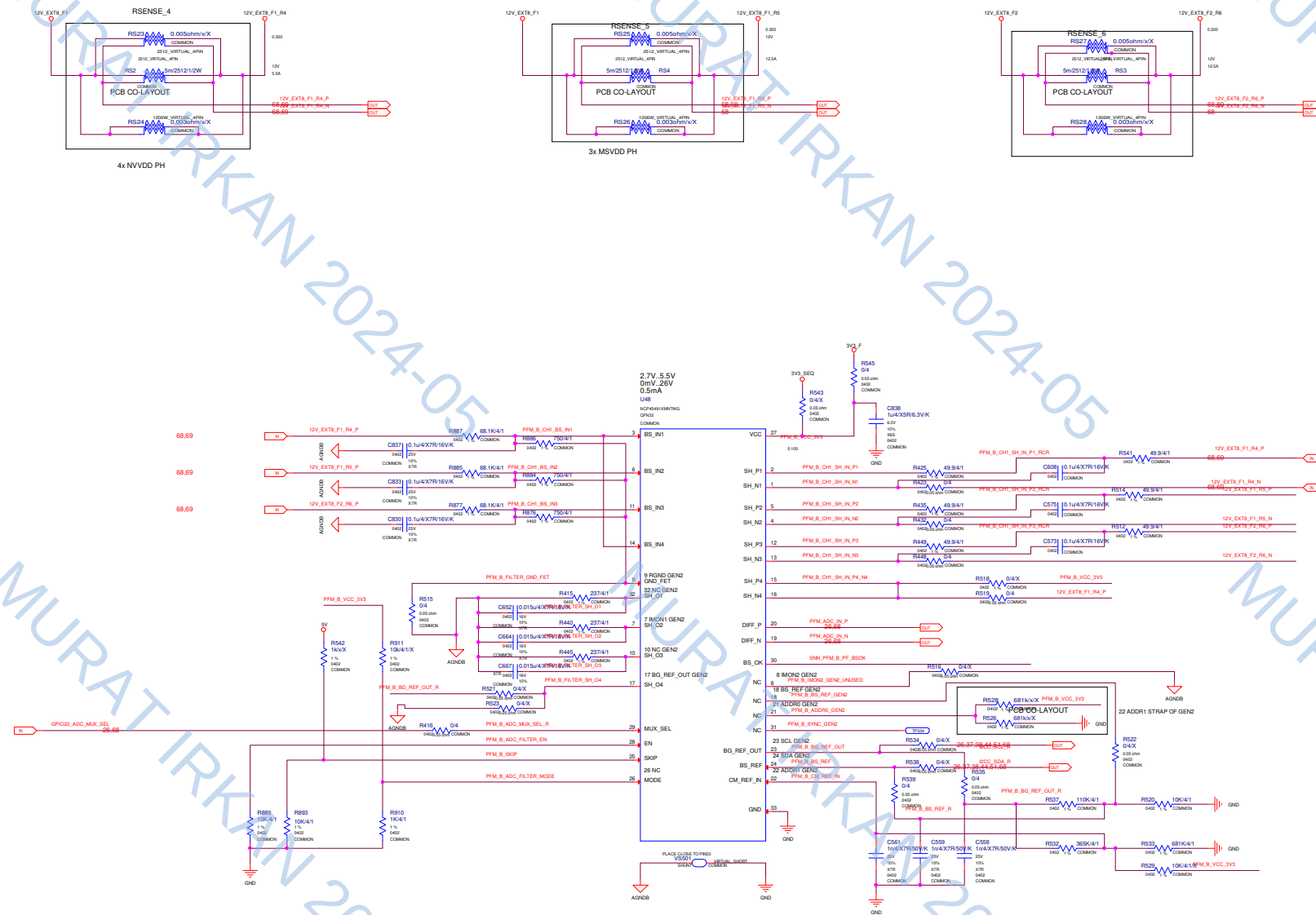
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0mV .26V
0.5mA
NCP469VANTING
07010
COMMONPLACE CLOSE TO PEX3
VDD1
COMMONASSEMBLY
PAGE DETAILASSEMBLY DESCRIPTION
PS: PRE-FILTER

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PS: Pre-Filter B

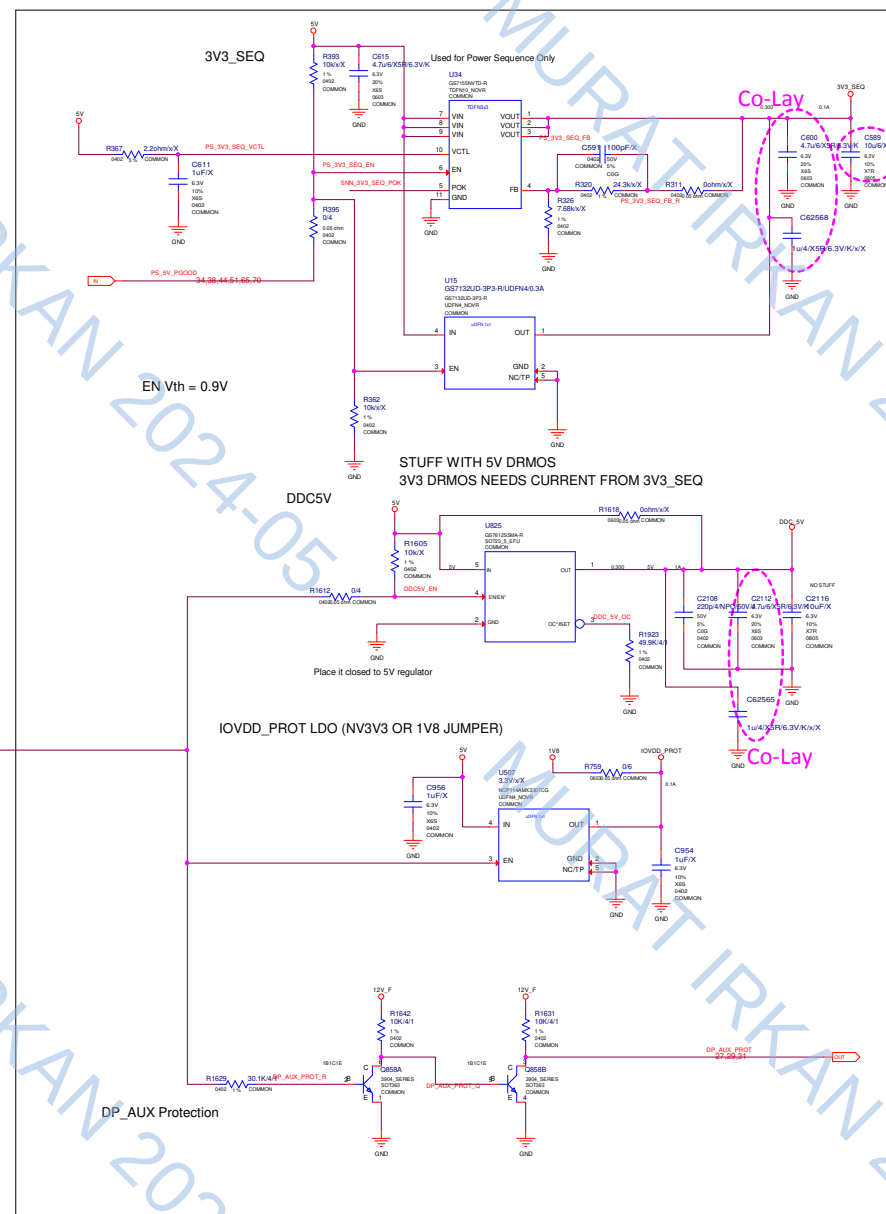
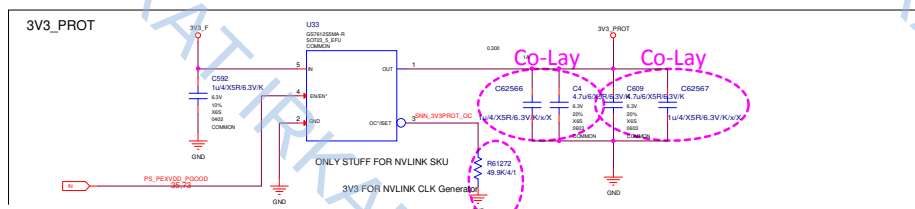
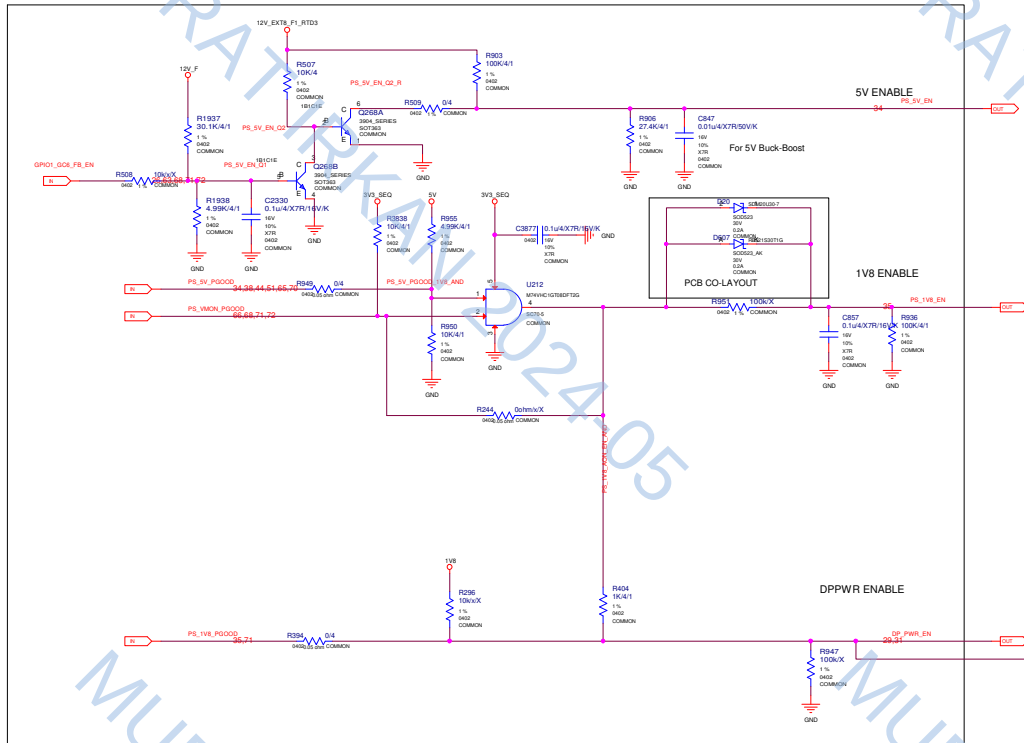


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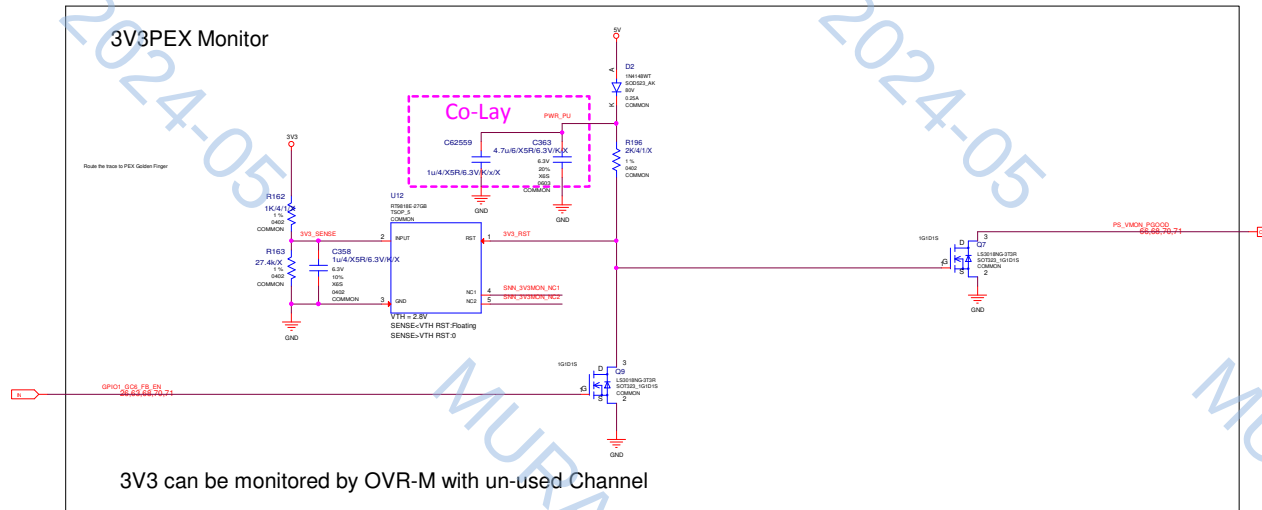
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Title: PCIE			
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SEQUENCE:5V,1V8,3V3_SEQ ENABLE



Sequence: 3V3 MONITOR



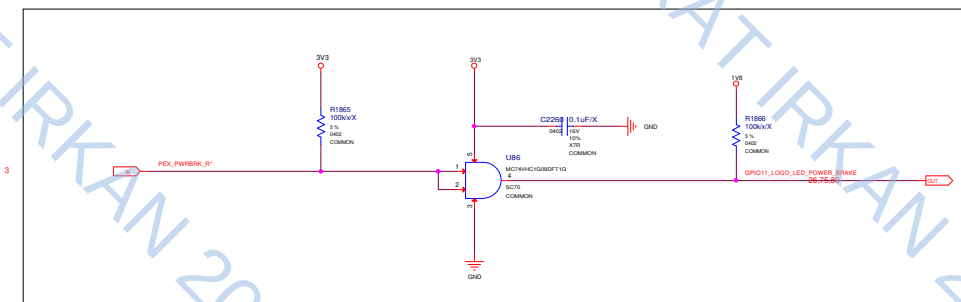
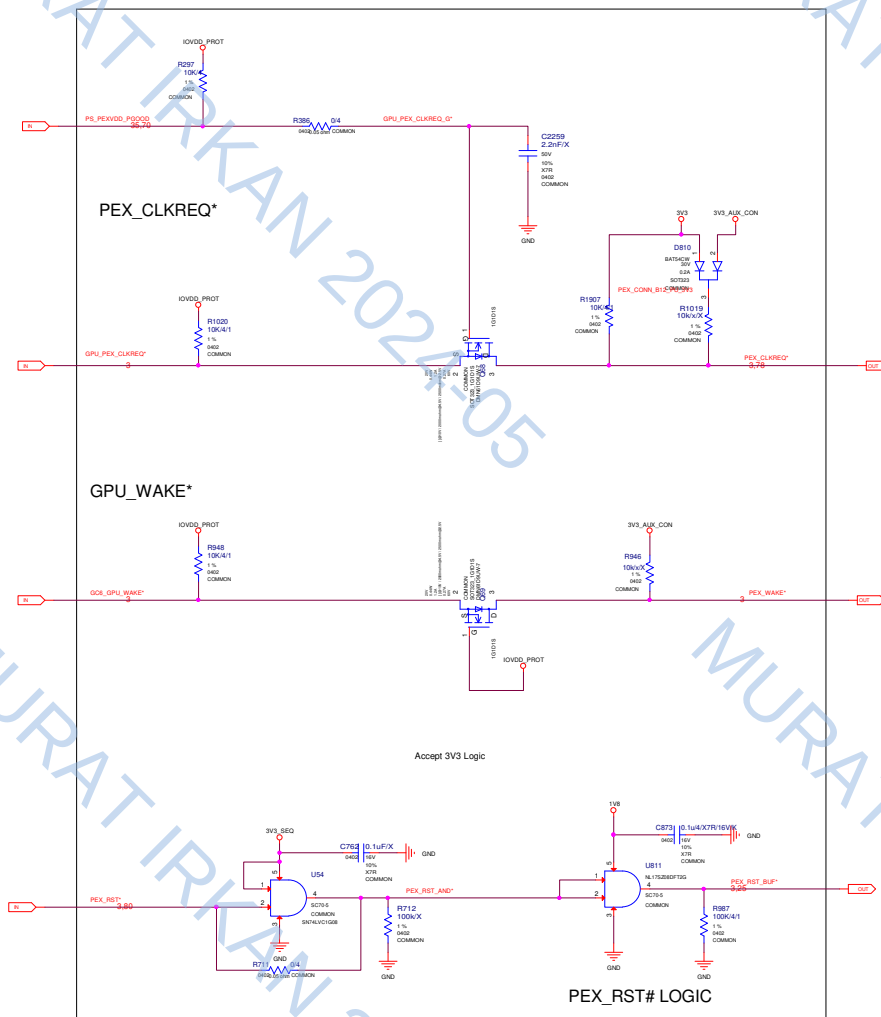
3V3 can be monitored by OVR-M with un-used Channel

GIGABYTE

Title			
PCIE			
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SEQUENCE:MISC



Accept 3V3 Logic

PEX_RST# LOGIC

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PAGE DETAIL

Sequence: MISC

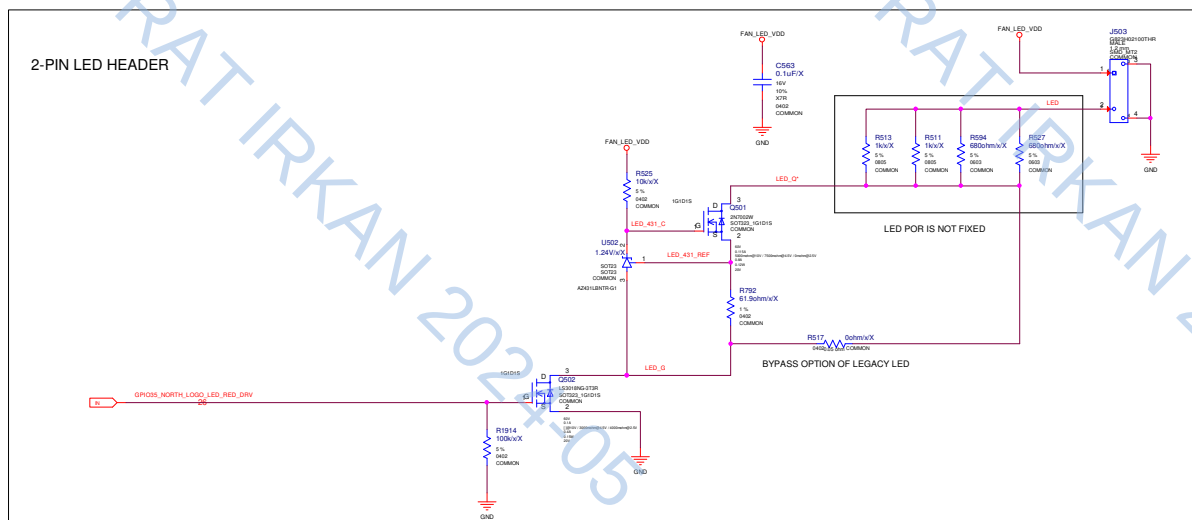
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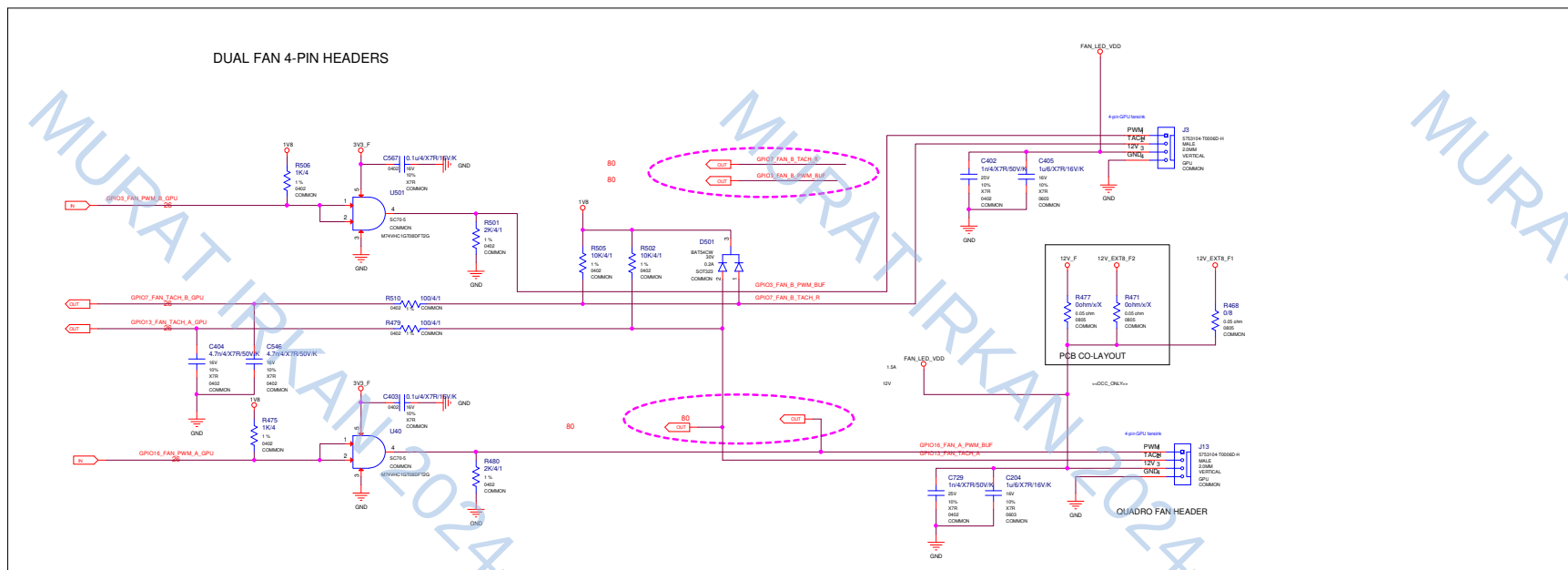
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MISC: LED & FAN

2-PIN LED HEADER



DUAL FAN 4-PIN HEADERS

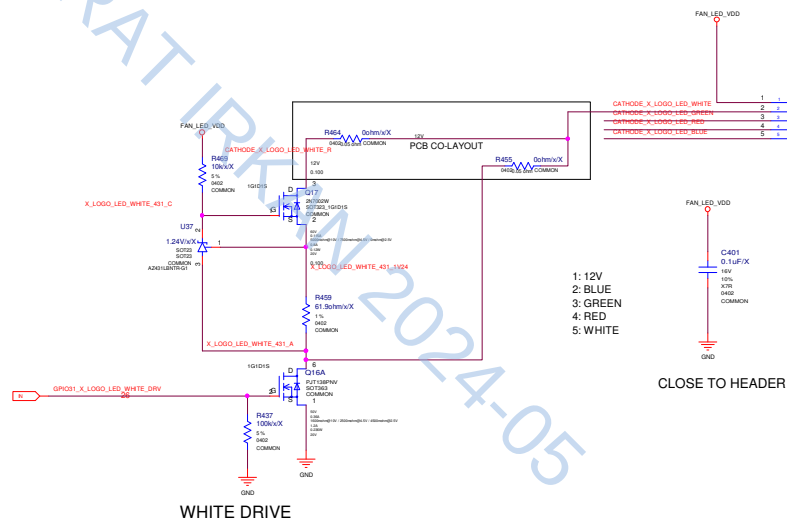


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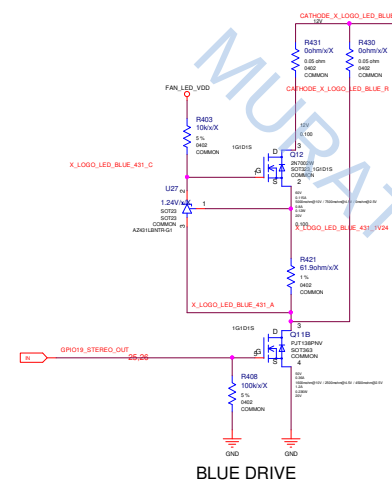
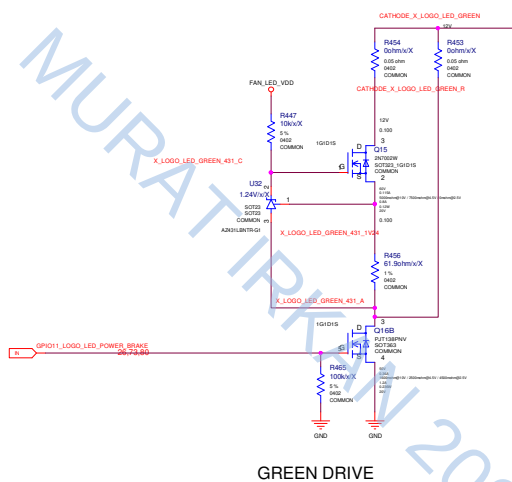
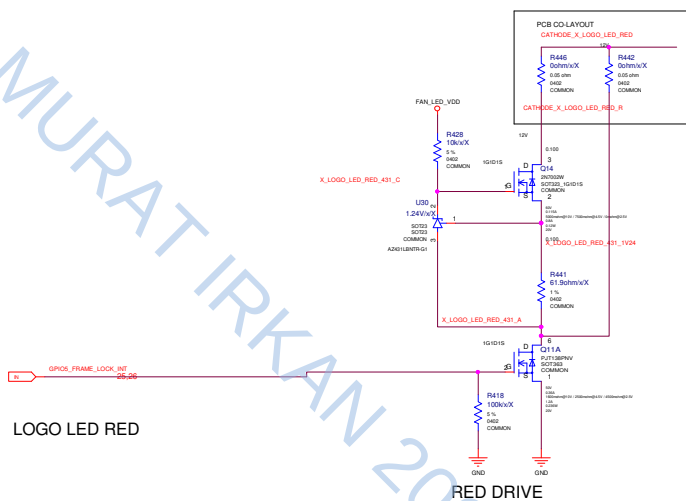
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CONNECTOR SIZE TO FIT LAYOUT
ALL CIRCUIT NO STUFF BY DEFAULT
ALINGN WITH PASH
IF ADDITIONAL POWER CONSUPTION R
NEEDED, IT SHOULD BE ON LED MOD.

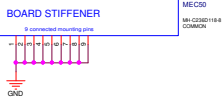


Mechanical: Mounting holes

BK up



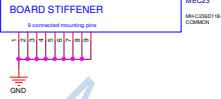
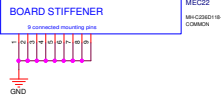
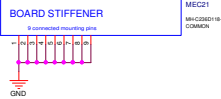
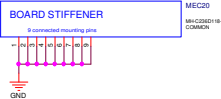
BK low



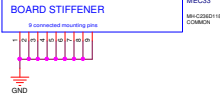
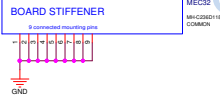
GPU



MOS



Backplane



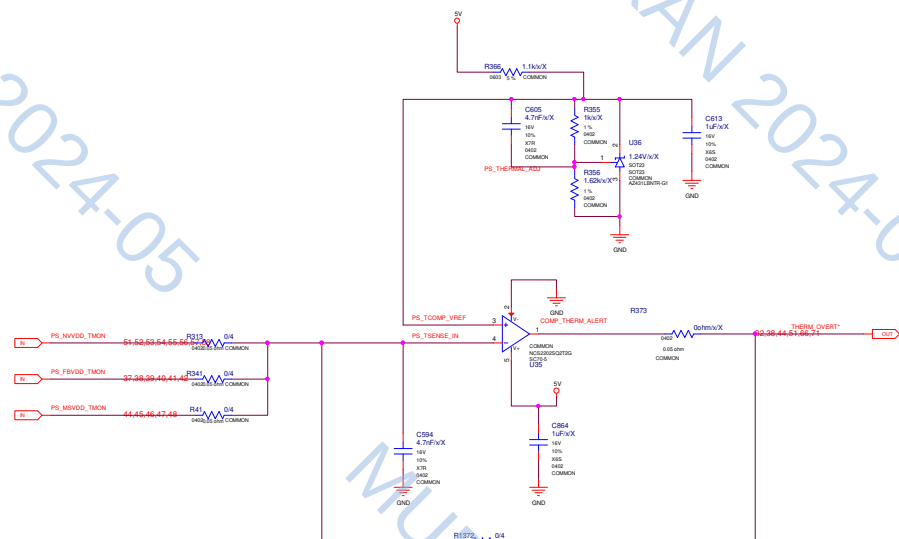
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VR THERMAL PROTECTION

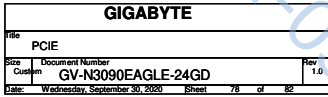


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VR Thermal Protection

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BIOS Switch

For Gaming or Server modes switch

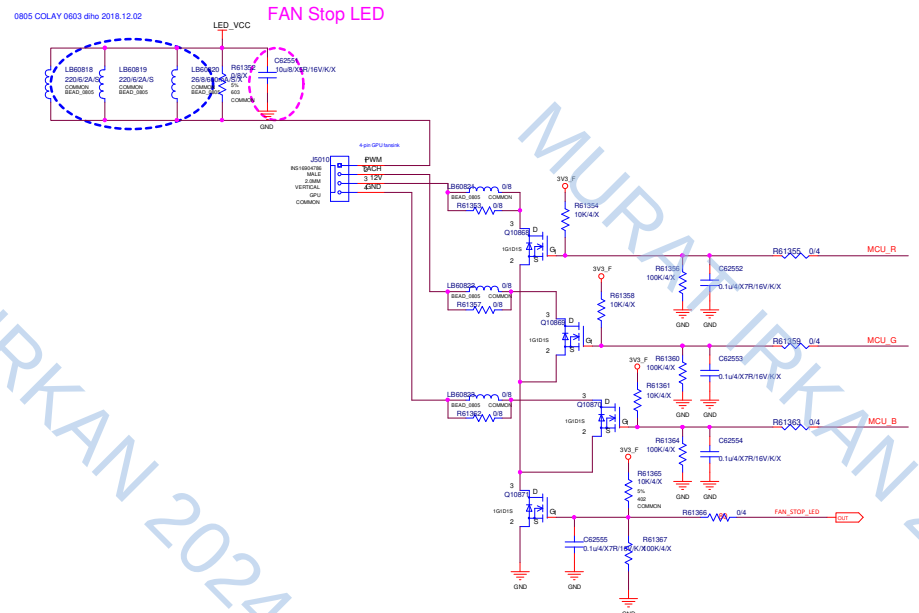
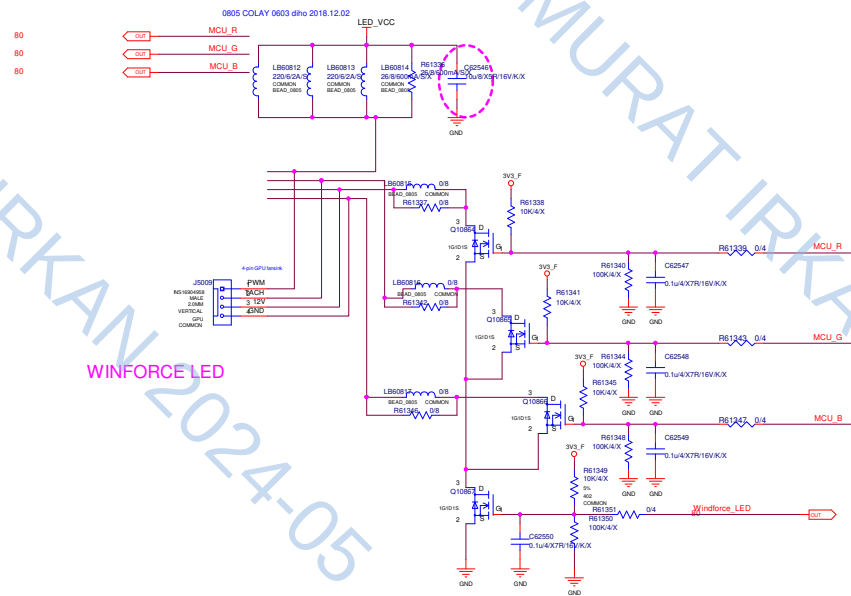
S	Function
0	B0 to A = DP2
1	B1 to A = DP1

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MCU ZERO POWER IDLE

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History

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2018.07.21	First version

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